Concept: Autonomous AI Platform with LINUX Client + Oracle VPS
Version 6.0 | Production State Integrated | For Students, Beginners, and Advanced Users | Maximum Version | Zero-Cost Software

Foreword
This concept describes a fully autonomous AI platform based on an Oracle Cloud server (Always Free) and an Omarchy laptop. It is written for students and beginners; every technical term is explained, and every step is justified. You set up Phases 0–3 manually; starting from Phase 4, an AI agent takes over.

Live Status (verified)
- **Client:** opencode 2.0.32, 26 MCP servers configured (21 active), 9 skills, 7 commands, 4 timers, voxtype/wake-word/openportal/rclone-rcd active.
- **VPS:** 30 containers, 0 unhealthy, 0 failed units, Disk 89%, 4 GB Swap, ufw + fail2ban active, Docker live-restore, journal 91 MB, TZ Europe/Berlin, 3 rclone remotes.

What you need
- Laptop (at least 8 GB RAM, 50 GB free storage; GPU optional, Vulkan recommended)
- Internet access
- Credit card for Oracle registration (will not be charged)
- Patience for one-time setup (approx. 4–5 hours incl. hardening)

What you get
- Free 24/7 server (Oracle Always Free)
- AI platform for photo/video analysis and automated social media posts
- Development environment with Git and CI/CD
- Voice control via Telegram **and** local voice stack (voxtype)
- Automatic backups and maintenance
- Local dictation in any application (Wayland native)
- Voice commands for system control
- Production-ready hardening: ufw, fail2ban, Caddy auth, secret scan, restore tests

What it costs
- 0 € for software
- 0 € for server
- 0 € for AI (except for cent amounts with heavy use)
- No domain needed
- Optional: ~10 €/year for your own domain

Target audience for this document
Part I contains cross-sectional chapters. **Chapters 1–2 are needed from Phase 0** (operating model, secrets). **Chapters 4–12 can be skipped on the first run** and are needed from Phase 11. Part II (Phases 0–14) is the entry point; Appendix F explains basic terms; Appendix H provides a learning path.

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PART I – CROSS-SECTIONAL CHAPTERS

1. Operations and Access Model
1.1 Task Distribution
Machine	Role	Services
VPS (Oracle)	24/7 Services	n8n, Forgejo, Postiz, Supabase, Temporal, Monitoring, Backups, Telegram Bot, Whisper Server
Laptop (Omarchy)	GPU/Interactive Loads	local STT/TTS, dictation, development, OpenCode work, voxtype, openportal

No load balancer, no cluster. Single-node with clear responsibilities and failover via backup/restore.

1.2 Access Order
1. Tailnet direct (preferred): `ssh oracle-vps`, `http://oracle-vps:8443`
2. `tailscale serve` for 127.0.0.1 services (e.g., Forgejo 3000)
3. Funnel only for true public access (max. 3 ports/node)

1.3 Firewall Reality
- Only 22/tcp is publicly reachable (SSH, key-only).
- Everything else is blocked via VCN Security List, `ufw`, and `DOCKER-USER`.
- Docker published ports bypass ufw → the `DOCKER-USER` chain is the actual boundary.
- `cloudflared` is inactive in the production state, no domain needed.
- If you want public access, use Tailscale Funnel – not both in parallel.

1.4 Acceptance Test
```bash
curl -s -o /dev/null -w "%{http_code}" http://<public-ip>:4007 # 000
curl -s -o /dev/null -w "%{http_code}" http://<public-ip>:8000 # 000
curl -s -o /dev/null -w "%{http_code}" http://100.90.252.42:8443 # 401
```

---

2. Secrets and Key Lifecycle
2.1 Storage
- No keys in `~/.bashrc`.
- Central file: `~/.config/opencode/env.sh` with `chmod 600`.
- Sourced by shell and every non-interactive start path (Hyprland keybind, systemd service, bot).
- In `opencode.json`, exclusively use `{env:VAR}` - no inline keys.
- Test interpolation live after the rebuild.

2.2 Secret Scan before every Cloud Upload
Patterns: `sk-`, `ghp_`, `github_pat_`, `AIza`, `eyJ...`, Private Keys. Hit = Abort, not "continue".

```bash
tar -tzf ~/backups/opencode-config-*.tar.zst | grep -E 'env\.sh|\.env$' # must be empty
```

2.3 Rules
- Secrets never in normal backups, never in logs, never in the frontend, never in chat output. Only location references.
- After a leak: rotate, don't just delete.
- Store recovery keys (`crypt.key`, `backup-secrets.key`) encrypted externally (7z AES-256), passphrase separate from the carrier (email without passphrase).

2.4 env.sh Template (one-time, applies to all phases)
```bash
mkdir -p ~/.config/opencode
cat > ~/.config/opencode/env.sh << 'ENV_EOF'
# Cloudflare (optional)
export CLOUDFLARE_API_TOKEN=""
export CLOUDFLARE_ACCOUNT_ID=""
export CLOUDFLARE_TUNNEL_TOKEN=""

# AI-Provider
export OPENROUTER_API_KEY=""
export NVIDIA_API_KEY=""
export GEMINI_API_KEY=""
export GROQ_API_KEY=""
export ZAI_API_KEY=""

# Cloud-Services
export SUPABASE_URL=""
export SUPABASE_ANON_KEY=""
export SUPABASE_SERVICE_ROLE_KEY=""
export VERCEL_TOKEN=""
export GITHUB_PERSONAL_ACCESS_TOKEN=""

# Resend
export RESEND_API_KEY=""
export SENDER_EMAIL_ADDRESS=""

# Telegram
export TELEGRAM_BOT_TOKEN=""
export TELEGRAM_ADMIN_USER_ID=""

# Forgejo (Phase 6)
export FORGEJO_READ_TOKEN=""
export FORGEJO_WRITE_TOKEN=""

# PDFgen (Phase 14)
export PDFGEN_API_KEY=""
```

Voice
export YDOTOOL_SOCKET=/tmp/.ydotool_socket
ENV_EOF
chmod 600 ~/.config/opencode/env.sh
```
In `.bashrc`:
```bash
[ -f ~/.config/opencode/env.sh ] && source ~/.config/opencode/env.sh
```

---

3. Backup & Disaster Recovery
3.1 3-2-1 Rule
- Local (Laptop/VPS)
- VPS cross-copy
- Encrypted offsite (rclone crypt → Google Drive)

3.2 Three Separate Backups
Backup	Content	Format	Retention
Projects + Configs	without `node_modules/.git/.venv`, without secrets	`tar.zst`	7 days normal
Databases	`pg_dump`	SQL.gz	7 days
Secrets separate	`env.sh`, `crypt.key`, `backup-secrets.key`	`gpg AES-256/SHA-512`	30 days local, 90 days offsite

3.3 Retention by Date, not by Name
- 7 days normal
- 30 days Secrets local
- 90 days offsite

3.4 Actually Test Restore
```bash
tar -xzf backup.tar.zst -C /tmp/restore-test
cmp /tmp/restore-test/file /original/file # 0 = identical
```
Test cross-host restore, because crypt remotes can point to different folders per host.

3.5 Document
Without `crypt.key` and `backup-secrets.key`, there is no restore. Keep both offline.

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4. Monitoring & Alerting
4.1 Stack (Reference for all Phases)
| Tool | Purpose | Port (Caddy) |
```

|---|---|---|
Uptime Kuma	Reachability	8443
Beszel (Hub + Agent)	CPU/RAM/Disk/Container Health	8444
Dozzle	Logs	8445
cAdvisor	Metrics	8446
Diun	New Image Versions	-
Spotlight	Dashboard	8447
monitor.sh	Status line via Telegram every 30 min	-

4.2 Alarms with Thresholds
- Status: 3 min
- CPU: 90% / 5 min
- RAM: 90% / 5 min
- Disk: 90% / 10 min
- Container Health
- Failed systemd units
Every alarm must actually arrive in Telegram (send a test message, don't assume).

4.3 update-check.timer
Weekly: only check and notify, do not install automatically. Commands:
- VPS: `apt list --upgradable`
- Client: `checkupdates` (pacman-contrib)
Output via Telegram.

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5. Reverse Proxy & UI Auth
5.1 Caddy as Tailnet-only Reverse Proxy
- Ports 8443-8447
- Basic Auth with bcrypt for Kuma (8443), Beszel (8444), Dozzle (8445), cAdvisor (8446), Spotlight (8447), n8n, Temporal UI
- Acceptance test: without auth 401, from outside 000

5.2 Postiz
- `DISABLE_REGISTRATION=true`
- Own 48-character JWT_SECRET, no default secret

5.3 Caddy Pitfalls
- `host:port` becomes implicitly HTTPS (400 "HTTP request to an HTTPS server") → `http://host:port` explicitly
- `basic_auth {...}` multiline

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6. Host Hardening
The following measures are executed once in Phase 1.4. Here is the reasoning:
- **ufw:** Default deny incoming, allow 22/tcp, `tailscale0`, 41641/udp. `DEFAULT_FORWARD_POLICY=ACCEPT`, otherwise Docker-FORWARD breaks.

- **fail2ban:** Brute-force protection for SSH.
- **SSH key-only:** No password login.
- **Docker:** `live-restore: true`, `userland-proxy: false`, journaled logging with 10 MB × 3.
- **journal:** `SystemMaxUse=100M`.
- **Swap:** 4 GB + `vm.swappiness=10` (Ampere otherwise has OOM risk).
- **Timezone:** `Europe/Berlin` system-wide (Ubuntu cron doesn't know `CRON_TZ`).
- **No Volume-Prune:** Only Image/Builder-Prune per timer.

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7. Update, Rollback, and Change Discipline
- No auto-update on the production stack. Diun only notifies; decisions and testing are manual.
- Before every change, backup: `.bak-<date>` or `.bak-<reason>-<date>`.
- Always consider rollback: which file, which service, which command.
- Verify change first (`systemctl status`, `docker ps`, `curl`, `ss -lnt`), then report as done.
- "Installed ≠ active" – check status/compare after every installation.

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8. Capacity, Quota, and Cost Guard
- Oracle Reclamation: <20% in the 95th percentile over 7 days (CPU + Net + RAM).
- Keepalive with real CPU load, 25% Duty (60 s on / 180 s off). 25% Duty generates significantly >20% in the 95th percentile because the load peaks pull the percentile. A pure idle instance would fall <20%.
- Disk Guard: Alarm at 90%, currently 89% → prepare Block-Volume migration (Always Free = 200 GB total).
- RAM Limits per Container: Postiz 1.5 GB (live 85% utilized – monitor).
- Know Free-Tier limits: Groq TPM/RPD, OpenRouter RPD, Gemini blocked, Resend without domain only `onboarding@resend.dev` to owner address.
- Don't assume "unlimited" – read headers live.

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9. Documentation and Memory Layer
File	Content
`PORTS.md`	Port inventory (Public/Tailnet/Loopback)
`SYSTEM.md`	Task distribution
`BACKUP.md`	Retention/Restore
`AGENTS.md`	per project

Memory structure:
```text
memory/
  index.md
  conventions.md
  projects/<name>.md
```

```
  sessions/<date>.md
  changelog.md
```
Append-only, backup before content changes. Maintain project register (path, stack, status). Primary source beats memory prose: Code/Git/Manifest/README before memory prose.

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10. Acceptance and Operations Checklist
The complete acceptance criteria are in Part IV. They are part of every phase, not just "at the end".

11. Incident Runbook
Escalation path: first status/logs, then backup, then intervention – never delete blindly.

Problem	First Steps
Container down	`docker ps -a`, `docker logs <name> --tail 100`, `docker compose up -d`
Disk full	`du -sh /*`, `docker system df`, image prune, check block volume
DB gone	check `pg_dump` backup, test restore, check volume mount
401/403	check token in `env.sh`, test `{env:}` interpolation
429	read rate-limit headers, switch provider (fallback)
OOM	`dmesg | grep -i oom`, increase RAM limit, check swap
Certificate	Caddy logs, renew `tailscale cert`
Tunnel dead	`systemctl status cloudflared`, `tailscale serve status`
Bot silent	`systemctl --user status opencode-telegram-bot`, check token
Backup broken	`rclone check`, `cmp` after restore, check `crypt.key`

12. Data Protection, Law, Licenses
- Document data flow: what stays local, what goes to OpenRouter/NVIDIA/Google/Groq/Cloudflare/Resend.
- GDPR: personal data only in services with an appropriate legal basis; encrypted backups; deletion periods.
- Licenses: AGPL (Postiz), MIT/Apache for tools – check commercial use.
- No secrets in frontend/logs; rotation after leak.

PART II – PHASES 0–14

PHASE 0 – Preparation (Human, ~60 Min)
0.1 What is a Phase?
A phase is a section in the workflow. You do Phases 0–3 yourself by hand. From Phase 4 onwards, an AI agent takes over. You give it prompts. A prompt is an instruction to the AI in plain English.

0.2 Create Accounts
| Account | URL | What for? | Credit Card? |

|---|---|---|---|
Oracle Cloud	cloud.oracle.com	Server	Yes (Verification)
Tailscale	tailscale.com	Mesh VPN	No
OpenRouter	openrouter.ai	AI Models	No
NVIDIA NIM	build.nvidia.com	AI Models	No
Google AI Studio	aistudio.google.com	AI Models	No
Groq	console.groq.com	AI + STT	No
z.ai	z.ai	AI Models	No
Supabase	supabase.com	Database	No
Vercel	vercel.com	Websites	No
GitHub	github.com	Code Management	No
Google Cloud	console.cloud.google.com	Google Drive	No
Telegram	Install App	Control	No
Resend	resend.com	E-Mail	No
LM Studio / Ollama	local	Offline Models	No
Forgejo	self-hosted	Git (no account)	No

No domain needed – Tailnet + Funnel are sufficient. Cloudflare is optional, not mandatory.

0.3 Collect Keys
Keys in `~/.config/opencode/env.sh` (chmod 600), not in `~/.bashrc`.
- **Cloudflare (optional):** API-Token + Tunnel-Token.
- **OpenRouter:** API-Key, Free-Tier 20 RPM / 50 RPD.
- **NVIDIA NIM:** Phone verification, copy key.
- **Google AI Studio:** Get API Key.
- **Groq:** API-Key (Free-Tier: 8000 TPM – unusable as main agent, only as Small Model).
- **z.ai:** API-Key (free on z.ai).
- **Supabase:** ANON_KEY, SERVICE_ROLE_KEY, URL.
- **Vercel:** Token.
- **GitHub PAT:** Contents (R/W), PRs (R/W), Issues (R/W), Actions (R), Metadata (R), Workflows (R/W).
- **Telegram:** @userinfobot → User-ID; @BotFather → Token.
- **Resend:** API-Key (Free: 3,000/month, without domain only `onboarding@resend.dev` to owner).
- **PDFgen:** API-Key on pdfgen.dev (Free-Tier with Watermark).
- **Google Drive API:** OAuth client ID (Desktop app).

0.4 Create env.sh
See Section 2.4. Enter the collected keys there.

0.5 Resend Restriction
Without a verified domain: only `onboarding@resend.dev`, only to owner address.

0.6 Oracle Credit Card
Always Free, but capacity check; plan for ARM shortage.

PHASE 1 – Bootstrap Oracle VPS (Human, ~20 Min)
1.1 Create VPS
1. Oracle Console → Compute → Instances → Create Instance
2. Name: `oracle-vps`
3. Image: Ubuntu 24.04 (ARM64)
4. Shape: `VM.Standard.A1.Flex` – 2 OCPUs, 12 GB RAM
5. Boot Volume: 100 GB (Always Free: 200 GB total)
6. SSH-Key: generate/upload
7. Networking: VCN with public subnet

If "Out of host capacity": Try another Availability Domain (`oci-arm-catcher`).

1.2 Bootstrap-Script
```bash
cat > ~/bootstrap.sh << 'BOOTSTRAP_EOF'
#!/bin/bash
set -e

echo "=== System Update ==="
sudo apt update && sudo apt upgrade -y

echo "=== Docker (ARM64) ==="
sudo apt install -y ca-certificates curl gnupg lsb-release
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
echo "deb [arch=arm64 signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
sudo usermod -aG docker $USER

echo "=== Node.js 22 ==="
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs

echo "=== Python + Build-Tools ==="
sudo apt install -y python3 python3-pip python3-venv python3-full git build-essential cmake
tmux rclone fail2ban

echo "=== uv ==="
curl -LsSf https://astral.sh/uv/install.sh | sh

echo "=== Tailscale ==="
curl -fsSL https://tailscale.com/install.sh | sh
```

```
echo "=== Directories ==="
mkdir -p ~/projects ~/.config/opencode

echo "=== Done - please log in again (docker group) ==="
BOOTSTRAP_EOF
chmod +x ~/bootstrap.sh
~/bootstrap.sh
```
Log out and back in:
```bash
exit
ssh -i /path/key ubuntu@YOUR_IP
```

1.3 Tailscale
```bash
sudo tailscale up --ssh --hostname=oracle-vps --accept-dns=true
tailscale status
```

1.4 Host Hardening
```bash
sudo ufw default deny incoming
sudo ufw allow 22/tcp
sudo ufw allow in on tailscale0
sudo ufw allow 41641/udp
sudo sed -i
's/DEFAULT_FORWARD_POLICY="DROP"/DEFAULT_FORWARD_POLICY="ACCEPT"/'
/etc/default/ufw
sudo ufw --force enable

sudo systemctl enable --now fail2ban

sudo sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/'
/etc/ssh/sshd_config
sudo systemctl restart ssh

sudo tee /etc/docker/daemon.json << 'EOF'
{
  "live-restore": true,
  "userland-proxy": false,
  "log-driver": "journald",
  "log-opts": {"max-size": "10m", "max-file": "3"}
}
EOF
sudo systemctl restart docker

echo 'SystemMaxUse=100M' | sudo tee -a /etc/systemd/journald.conf
```

sudo systemctl restart systemd-journald

sudo fallocate -l 4G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
echo 'vm.swappiness=10' | sudo tee -a /etc/sysctl.conf
sudo sysctl -p

sudo timedatectl set-timezone Europe/Berlin
```

### 1.5 Create PORTS.md
```markdown
# PORTS.md
| Service | Public | Tailnet | Loopback | Docker-internal |
|---|---|---|---|---|
| SSH | 22/tcp | - | - | - |
| Forgejo | - | 3000 (serve) | 127.0.0.1:3000 | - |
| n8n | - | 8443 (Caddy) | 127.0.0.1:5678 | - |
| Postiz | - | 8443 (Caddy) | 127.0.0.1:4007 | - |
| Temporal UI | - | 8443 (Caddy) | 127.0.0.1:8080 | 7233 |
| Supabase | - | 8443 (Caddy) | 127.0.0.1:8000 | 5432/6543 |
| Whisper | - | - | 127.0.0.1:8081 | - |
| rclone RC | - | - | 127.0.0.1:5572 | - |
| Uptime Kuma | - | 8443 (Caddy) | 127.0.0.1:3001 | - |
| Beszel | - | 8444 (Caddy) | 127.0.0.1:8090 | - |
| Dozzle | - | 8445 (Caddy) | 127.0.0.1:8082 | - |
| cAdvisor | - | 8446 (Caddy) | 127.0.0.1:8083 | - |
| Spotlight | - | 8447 (Caddy) | 127.0.0.1:8084 | - |
| Woodpecker (optional) | - | 8443 (Caddy) | 127.0.0.1:8002 | - |
```
Total: 30 containers (incl. Supabase stack, Temporal, Postiz, n8n, Forgejo, Monitoring, Whisper, Piper, OpenClaw, rclone, Caddy). Woodpecker uses 8002 to avoid collision with Supabase (8000).

1.6 Block-Volume Plan
Always Free = 200 GB total. Move `/var/lib/docker` and `/var/lib/containerd` if Disk >90%.

1.7 Verify
```bash
docker --version
node --version
opencode --version
tailscale status
```
```

PHASE 2 – Bootstrap Laptop (Human, ~30 Min)
2.1 Install Omarchy
1. Omarchy ISO from omarchy.org
2. USB: `sudo dd if=omarchy.iso of=/dev/sdX bs=4M status=progress oflag=sync`
3. Boot from USB → Installer
4. After Reboot: Omarchy Configurator

2.2 Base Setup
```bash
cat > ~/bootstrap.sh << 'BOOTSTRAP_EOF'
#!/bin/bash
set -e

echo "=== System Update ==="
sudo pacman -Syu --noconfirm

echo "=== Base Packages ==="
sudo pacman -S --noconfirm --needed \
  nodejs npm python python-pip python-pipx \
  git docker docker-compose tmux rclone \
  base-devel openssl zlib xz tk zstd cmake \
  7zip github-cli git-delta pacman-contrib

echo "=== Activate Docker ==="
sudo systemctl enable --now docker
sudo usermod -aG docker $USER

echo "=== uv ==="
curl -LsSf https://astral.sh/uv/install.sh | sh

echo "=== OpenCode ==="
sudo npm install -g opencode-ai opencode-auto-fallback opencode-live

echo "=== voxtype ==="
# yay is preinstalled on Omarchy; if not:
# sudo pacman -S --needed --noconfirm base-devel git
# git clone https://aur.archlinux.org/yay.git /tmp/yay && (cd /tmp/yay && makepkg -si
--noconfirm)
yay -S --noconfirm voxtype-bin

echo "=== Freebuff (passive) ==="
sudo npm install -g freebuff

echo "=== Directories ==="
mkdir -p ~/projects ~/Work ~/.config/opencode
```

````
echo "=== Done - please log in again ==="
BOOTSTRAP_EOF
chmod +x ~/bootstrap.sh
~/bootstrap.sh
```

### 2.3 Tailscale
```bash
sudo pacman -S --noconfirm tailscale
sudo systemctl enable --now tailscaled
sudo tailscale up --ssh --hostname=laptop --accept-dns=true
tailscale status
ssh oracle-vps
```

### 2.4 SSH Config
`~/.ssh/config`:
```
Host oracle-vps
  HostName oracle-vps
  User ubuntu
  ForwardAgent yes
```

### 2.5 Set up Timers
- git-sync 22:00
- projects-backup 04:00
- secrets-backup 04:30
- update-check Mon 08:00 (`checkupdates`)

### 2.6 rclone-rcd.service (Laptop, User-Service)
```ini
# ~/.config/systemd/user/rclone-rcd.service
[Unit]
Description=Rclone RC Daemon
After=network.target

[Service]
ExecStart=/usr/bin/rclone rcd --rc-no-auth --rc-addr=127.0.0.1:5572
Restart=always
RestartSec=10

[Install]
WantedBy=default.target
```
```bash
systemctl --user enable --now rclone-rcd
```
````
````

2.7 Additional Client Services
- `omarchy-wake-word.service`
- `omarchy-crash-watch.service`
- `voxtype` (see Phase 9.2)
- `openportal` (see Phase 8.3)
- `rclone-rcd` (see 2.6)

2.8 mise/Node Paths, pxexec, AUR
- mise for tool versions
- pxexec pattern for installations without TTY
- AUR: `7zip`, `github-cli`, `git-delta`

PHASE 3 – OpenCode + AI Provider (Human, ~15 Min)
3.1 OpenCode
Terminal-based AI agent. Programs, edits files, executes commands.

3.2 Provider Strategy (8 Providers)
Priority	Model	Provider	Context	Free-Tier
Primary	Groq gpt-oss-120b	Groq	-	8000 TPM (only Small Model)
Secondary	Nemotron 3 Ultra Free	OpenRouter	1M	50 RPD
Tertiary	Nemotron 3.5 Lightning	NVIDIA NIM	262K	40 RPM
Quaternary	Gemini 2.5 Flash-Lite	Google AI Studio	1M	1,500 RPD
Small	Groq gpt-oss-120b	Groq	-	-
Local	qwen3:8b, qwen3-vl:8b	LM Studio/Ollama	-	0 €
Optional	GLM-4.5-Flash	z.ai	-	free

OpenCode Zen is no longer the main path and is deliberately not configured
(`disabled_providers` is not set).

3.3 opencode.json (global)
```bash
mkdir -p ~/.config/opencode
cat > ~/.config/opencode/opencode.json << 'EOF'
{
  "$schema": "https://opencode.ai/config.json",
  "plugin": ["opencode-live", "opencode-auto-fallback"],
  "provider": {
    "groq": {
      "type": "openai",
      "baseURL": "https://api.groq.com/openai/v1",
      "apiKey": "{env:GROQ_API_KEY}",
      "models": {
        "gpt-oss-120b": { "name": "Groq GPT-OSS 120B" }
      }
```

```
    },
    "openrouter": {
      "type": "openai",
      "baseURL": "https://openrouter.ai/api/v1",
      "apiKey": "{env:OPENROUTER_API_KEY}",
      "models": {
        "nvidia/nemotron-3-ultra-550b-a55b:free": { "name": "Nemotron 3 Ultra Free" }
      }
    },
    "nvidia": {
      "type": "openai",
      "baseURL": "https://integrate.api.nvidia.com/v1",
      "apiKey": "{env:NVIDIA_API_KEY}",
      "models": {
        "nvidia/nemotron-3.5-lightning-30b-a3b": { "name": "Nemotron 3.5 Lightning" }
      }
    },
    "google": {
      "type": "google",
      "baseURL": "https://generativelanguage.googleapis.com/v1beta",
      "apiKey": "{env:GEMINI_API_KEY}",
      "models": {
        "gemini-2.5-flash-lite": { "name": "Gemini 2.5 Flash-Lite" }
      }
    },
    "lmstudio": {
      "type": "openai",
      "baseURL": "http://127.0.0.1:1234/v1",
      "models": {
        "qwen3:8b": { "name": "Qwen3 8B (local)" }
      }
    },
    "zai": {
      "type": "openai",
      "baseURL": "https://api.z.ai/v1",
      "apiKey": "{env:ZAI_API_KEY}",
      "models": {
        "glm-4.5-flash": { "name": "GLM-4.5-Flash" }
      }
    }
  },
  "model": "groq/gpt-oss-120b",
  "small_model": "groq/gpt-oss-120b"
}
EOF
```

3.4 Config Split

- Global `opencode.json` = Provider/Definitions
- Project config = Enabled mask
- No provider block in project

3.5 Test `{env:}` Interpolation
```bash
echo $GROQ_API_KEY
echo $OPENROUTER_API_KEY
curl -s https://api.groq.com/openai/v1/models -H "Authorization: Bearer $GROQ_API_KEY" | head -20
opencode models
```

3.6 Non-interactive Start Paths
Every start path (Hyprland keybind, systemd service, bot) must source `env.sh`.

3.7 Local Models
LM Studio + Ollama (`qwen3:8b`, `qwen3-vl:8b`) as offline fallback.

3.8 Additional Config Fields
- `plugins` object (Notifier)
- `skills.paths`
- `instructions`
- `agent` definitions (build/plan)
- `whitelist` / `disabled_providers`

PHASE 4 – Agent: Network & Zero Trust
4.1 Tailnet-first
Cloudflare is not mandatory. Tailnet-first:
- `tailscale serve --tcp` for loopback services
- MagicDNS for device names
- Grants for access rules
- Taildrop for file transfer

4.2 Caddy tailnet as Reverse Proxy (Installation)
Prompt:
> I want to install Caddy as a Tailnet-only reverse proxy.
> 1. Install Caddy as systemd service (pacman -S caddy on Arch, apt install caddy on Ubuntu).
> 2. Configure /etc/caddy/Caddyfile: - Ports 8443-8447 - Basic Auth with bcrypt for Kuma (8443), Beszel (8444), Dozzle (8445), cAdvisor (8446), Spotlight (8447), n8n, Temporal-UI - Reverse proxy to 127.0.0.1:<port>
> 3. Start: systemctl enable --now caddy
> 4. Check: systemctl status caddy
> Important: host:port becomes implicitly HTTPS; basic_auth multiline.

4.3 VCN Security List + DOCKER-USER
Docker published ports bypass ufw. The actual boundary is VCN + DOCKER-USER.

4.4 Acceptance Test
```bash
curl -s -o /dev/null -w "%{http_code}" http://<public-ip>:4007 # 000
curl -s -o /dev/null -w "%{http_code}" http://<public-ip>:8000 # 000
curl -s -o /dev/null -w "%{http_code}" http://100.90.252.42:8443 # 401
```

4.5 cloudflared deliberately inactive
No domain. Do not misunderstand as "still missing".

4.6 Funnel
Only for max. 3 ports/node. Funnel attribute in `nodeAttrs`.

4.7 Prompt: Tailscale Grants
> I want to migrate my Tailscale ACLs to Grants. 1. Open https://login.tailscale.com/admin/acls 2. Create a Grants policy with src/dst/ip rules for Laptop and VPS. 3. Check with tailscale status and tailscale ping oracle-vps. 4. Document the Grants in PORTS.md.

4.8 Optional Prompt: Cloudflare Tunnel
> I want to optionally set up Cloudflare Tunnel (cloudflared). 1. Check: cloudflared --version 2. Authenticate: cloudflared tunnel login 3. Create tunnel: cloudflared tunnel create oracle-vps 4. Create ~/.cloudflared/config.yml with ingress for postiz, git, n8n, woodpecker. 5. Route DNS for each hostname. 6. Install as systemd service. 7. Check: sudo systemctl status cloudflared. Ask for CLOUDFLARE_DOMAIN. Report reachable URLs.

4.9 Optional Prompt: Cloudflare MCP
> Add cloudflare-api, cloudflare-docs, cloudflare-bindings, cloudflare-builds, cloudflare-observability to ~/.config/opencode/opencode.json. Use {env:CLOUDFLARE_API_TOKEN}. Check JSON syntax and report available tools.

PHASE 5 – Agent: VPS Services
5.1 Real Stack: 30 Containers
- Temporal (7233, UI 8080)
- Supabase: Envoy/Kong (8000), Pooler (5432/6543), Studio, Storage, Auth, Realtime, Edge Functions, Imgproxy, Meta
- n8n (5678, Loopback + Caddy)
- Postiz (4007, 1.5 GB RAM limit, `DISABLE_REGISTRATION=true`, 48-char JWT)
- Forgejo (3000, Loopback + tailscale serve)
- Monitoring: Uptime Kuma, Dozzle, cAdvisor, Beszel, Diun, Spotlight
- Whisper-Server (8081)
- Piper TTS

5.2 Start Order
1. Temporal → Postiz (otherwise 502)
2. Supabase-DB → Auth/REST

5.3 Healthchecks + restart
Every service: `healthcheck` + `restart: unless-stopped`. Compose project isolation (otherwise "orphan").

5.4 Architecture: n8n, OpenClaw, Telegram Bot, OpenCode
n8n = Workflow orchestration (Triggers, Filters, Routing).
OpenClaw = Persistent media analysis agent + Gateway. Endpoint: `POST http://127.0.0.1:18789/analyze` with `{ "type": "image", "url": "..." }`.
Telegram Bot = User interface (Chat, Ops, Jobs).
OpenCode = Code agent (Files, Commands, Sessions).

5.5 Prompt: Postiz
> I want to install Postiz as a self-hosted social media scheduler. 1. Clone https://github.com/gitroomhq/postiz-docker-compose to ~/projects/postiz 2. Copy .env.example to .env 3. Set JWT_SECRET (48 chars), MAIN_URL, FRONTEND_URL, NEXT_PUBLIC_BACKEND_URL. 4. Set DISABLE_REGISTRATION=true. 5. Set RAM limit 1.5G. 6. Start docker compose up -d. 7. Wait 5 minutes, check docker compose ps and logs. Important: ARM64, start Temporal first. Report access URL.

5.6 Prompt: Supabase
> I want to install Supabase self-hosted. 1. Clone https://github.com/supabase/supabase (depth 1) to ~/projects/supabase 2. Change to supabase/docker 3. Copy .env.example to .env 4. Generate secure values for POSTGRES_PASSWORD, JWT_SECRET, ANON_KEY, SERVICE_ROLE_KEY 5. Start docker compose up -d 6. Wait 5 minutes 7. Check docker compose ps and curl http://localhost:8000

5.7 Prompt: n8n
> 1. Create ~/projects/n8n 2. docker-compose.yml: Image n8nio/n8n (ARM64), Port 5678, Volume n8n_data:/home/node/.n8n, restart unless-stopped, N8N_HOST, N8N_PORT, WEBHOOK_URL, N8N_ENFORCE_SETTINGS_FILE_PERMISSIONS=true. 3. Start docker compose up -d. 4. Check curl http://localhost:5678. 5. Bind n8n only to Loopback and Caddy/Basic Auth, not public.

5.8 Prompt: OpenClaw Gateway
> I want to install OpenClaw as a media analysis agent and gateway. 1. curl -fsSL https://openclaw.ai/install.sh | bash 2. source ~/.bashrc 3. On "How do you want to hatch your bot?" → "Do this later" 4. openclaw config set gateway.bind loopback 5. openclaw config set gateway.auth.mode token 6. openclaw doctor --generate-gateway-token 7. openclaw config set gateway.tailscale.mode serve 8. openclaw config set gateway.trustedProxies ["127.0.0.1"] 9. openclaw gateway install 10. systemctl --user restart openclaw-gateway.service 11. ARM: export NODE_COMPILE_CACHE=/var/tmp/openclaw-compile-cache; export OPENCLAW_NO_RESPAWN=1 12. Verification: openclaw --version; systemctl --user status openclaw-gateway.service; tailscale serve status; curl http://localhost:18789

5.9 Prompt: Social Media Workflow
> Build in n8n: 1. Google Drive Trigger: New File in Folder (Upload folder) 2. MIME-Filter: only image/* and video/* 3. Duplicate detection: persistent list (Redis or JSON) 4. HTTP Request to OpenClaw Gateway: Media analysis (POST http://127.0.0.1:18789/analyze) 5. Wait for analysis result 6. Switch task_type: image_analysis → OpenClaw Vision; post_creation → Postiz MCP; notify → Telegram; default → OpenClaw General 7. LLM text generation: Post text, hashtags, tags (style per account) 8. Postiz API: Create Draft (Status DRAFT) 9. Human-in-the-Loop: signed Approve/Reject links 10. Notification: Telegram + E-Mail in parallel 11. Wait for approval 12. Approve: Postiz DRAFT → QUEUE → Publish 13. Edit: Changes back, re-approval 14. Reject: Draft remains, notification. Document all nodes. Report webhook URLs and test end-to-end.

5.10 Prompt: Human-in-the-Loop
> Telegram Bot: 1. BotFather, Token 2. Token in n8n Credential 3. Chat-ID recipient 4. Message: Preview image, post text, hashtags, target platforms, inline buttons (Approve, Edit, Reject) 5. Callback to n8n Wait-Node. E-Mail: 1. SMTP Credentials (Gmail App-Password or Resend) 2. HTML with preview, signed links, link to Postiz UI. HITL-Node: 1. Install n8n-nodes-human-in-the-loop-custom 2. Signed Approve/Reject links (one-time, time-limited) 3. Timeout 24h → Telegram reminder, then E-Mail, then Draft. Test both channels.

5.11 Prompt: Platform-Specific Adaptation
> Configure in n8n: Instagram: Feed, Reel, Story, Carousel; max. 30 hashtags; tags; Alt-Text; 11-13 h, 19-21 h. TikTok: Video 9:16; Hook first 3 seconds; Trend sounds; Caption with keywords; no external links. YouTube Shorts: Video 9:16, max. 60 s; Title with keywords; Description; Tags. LinkedIn: Text + Image/Video; professional tone; fewer hashtags; Document-Carousel. X/Twitter: Text + Media; short, concise; Threads possible. Store as configuration and link with post generation.

5.12 Monitoring Containers
Uptime Kuma, Dozzle, cAdvisor, Beszel, Diun, Spotlight. Details in Section 4.1.

PHASE 6 – Agent: Git + CI/CD
6.1 Prompt: Forgejo
> 1. Create /opt/forgejo (sudo) 2. docker-compose.yml: Forgejo (codeberg.org/forgejo/forgejo:9), Postgres 16-alpine, Ports 3000 HTTP, 2222 SSH, Volumes forgejo_data, forgejo_db, ROOT_URL http://localhost:3000. 3. Secure DB passwords. 4. Start docker compose up -d. 5. Check curl http://localhost:3000. 6. Bind only 127.0.0.1:3000 + tailscale serve --tcp 3000 tcp://127.0.0.1:3000. Important: ARM64. Report URL.

6.2 Prompt: Forgejo Runner
> 1. Download Forgejo Runner ARM64: https://code.forgejo.org/forgejo/runner/releases/download/v6.1.0/forgejo-runner-6.1.0-linux-arm64 2. chmod +x, move to /usr/local/bin/forgejo-runner 3. Create

/etc/systemd/system/forgejo-runner.service 4. Activate service. Ask for registration token and instance URL after Forgejo installation.

6.3 Prompt: Woodpecker CI (optional)
> 1. /opt/woodpecker 2. docker-compose.yml: woodpecker-server Port 8002, woodpecker-agent connected to Forgejo, WOODPECKER_AGENT_SECRET, WOODPECKER_GITEA=true, WOODPECKER_GITEA_URL=http://forgejo:3000. 3. Start docker compose up -d. 4. Check curl http://localhost:8002. ARM64. Report URL.

6.4 Prompt: Forgejo MCP
> 1. npm install -g @ric_/forgejo-mcp 2. In ~/.config/opencode/opencode.json under mcp: "forgejo": { "type": "local", "command": "npm", "args": ["-y", "@ric_/forgejo-mcp"], "env": { "FORGEJO_URL": "http://localhost:3000", "FORGEJO_TOKEN": "{env:FORGEJO_READ_TOKEN}" }, "enabled": true } 3. Ask for FORGEJO_READ_TOKEN (Forgejo Settings → Applications → Generate Token) 4. Enter token in env.sh. Report tools.

6.5 Repo Hygiene
- Don't sync foreign clones (postiz/supabase/whisper.cpp)
- Only own repos with `origin`
- Set Git identity (`user.name`, `user.email`)

6.6 Token Reality
- VPS GitHub token 401 → no push from VPS
- Laptop uses `gh auth git-credential`

6.7 Worktrees
`worktree_create`/`worktree_list`/`worktree_remove`, `session_move`/`session_rename` for parallel work.

6.8 Forgejo MCP Read/Write
Read/Write tokens separated (`FORGEJO_READ_TOKEN`, `FORGEJO_WRITE_TOKEN`).

PHASE 7 – Agent: Telegram Bot & Voice Server
7.1 Bot v11 Architecture
- Scope router: `build` / `ops` / `chat`
- Deterministic Layer 0: fixed commands
- Groq gpt-oss-120b: primary
- NVIDIA escalation on Groq-429, Groq-5xx or tool-use error after 2 retries
- Deep path: via opencode

7.2 Job Engine
- One job per chat
- Max. 2 global
- 600 s timeout
- Progress edit every 5 s
- `/jobs`, `/log`, `/cancel`, `/cd`, `/pwd`

7.3 Target Switching
`/target vps|laptop`, `@vps`/`@laptop`, separate session IDs and workdirs.

7.4 Write Tools
`write_file`/`edit_file` only `/home/ubuntu` + `/tmp`. Backup before every change. Unique match on `edit_file`.

7.5 Destructive Guard
`rm`, `dd`, `mkfs`, `reboot`, `systemctl stop`, `docker rm` → only after `.ja` ("yes").

7.6 Uploads
`/home/ubuntu/inbox/<ts>_<name>`.

7.7 State
- `state.json` (0600)
- `qa.json` (0600)
- Facts-Refresh every 300 s

7.8 Prompt: Telegram Bot
> I want to install a Telegram bot. Package: @levra7/opencode-telegram-bot (npm) Features: Multi-User, SSH-Remote, Voice (Whisper), MCP-Visibility, Session-Management, Model-Switching, Agent-Modes. 1. npm install -g @levra7/opencode-telegram-bot 2. Setup-Wizard: Bot-Token (TELEGRAM_BOT_TOKEN), Telegram User-ID (TELEGRAM_ADMIN_USER_ID), OpenCode Server URL (http://localhost:4096) 3. Bot installs systemd service. 4. Check systemctl --user status opencode-telegram-bot. Prerequisite: opencode serve --hostname 127.0.0.1 --port 4096 &

7.9 Prompt: Whisper Server
> 1. Clone https://github.com/ggerganov/whisper.cpp to ~/projects/whisper.cpp 2. make 3. ./models/download-ggml-model.sh base 4. /etc/systemd/system/whisper-server.service: whisper-server -m models/ggml-base.bin --host 0.0.0.0 --port 8081 --language en 5. Activate service 6. Check curl http://localhost:8081. Important: ARM64 NEON.

7.10 Prompt: Piper TTS
> 1. Clone https://github.com/mevdschee/piper-voices-tts to ~/projects/piper-voices-tts 2. ./piper-voices.sh 3. Telegram bot: TTS_ENABLED=true, TTS_PROVIDER=piper, TTS_VOICE=en_US-amy-medium 4. Restart bot. Report TTS status.

7.11 STT Priority
1. Groq whisper-large-v3-turbo (0.3-0.6 s)
2. Local Whisper-Server 8081 as fallback if Groq 429/5xx or rate limit reached.
3. Bot logic: Groq first, local on error.

7.12 TTS: Piper
- `en_US-amy-medium`
- Opus 48 kHz

- `tts_clean` (Code blocks → "Code is in the text", Markdown/URLs/Emojis out, % → "percent", capped at 600 chars)
- Pronunciation map (`Bot` → `Bott`, `systemctl` → `System-Control`, `JSON` → `Jot Ess Oh Enn`)
- `/voice on|off`

7.13 Rate-Limiter + Error Handler
Against Telegram Bad-Gateway spam. `Linger=yes` for user services.

PHASE 8 – Agent: Full Session Control
8.1 Prompt: tmux + opencode-live
> On VPS: 1. Check tmux. 2. ~/tmux.conf: Prefix C-a, mouse active, base-index 1, split bindings | and -. 3. Copy same ~/tmux.conf to Laptop. opencode-live: 1. npm list -g opencode-live 2. If not: npm install -g opencode-live 3. Document in ~/projects/README-sessions.md how to start a shared session.

8.2 Prompt: Conduit + remote-ssh-mcp
> Conduit: 1. opencode serve --hostname 0.0.0.0 --port 4096 & 2. npx conduit-code 3. systemd service 4. Report URL. remote-ssh-mcp: 1. Install uv 2. uv tool install remote-ssh-mcp 3. In ~/.config/opencode/opencode.json under mcp: "remote-ssh": { "type": "local", "command": "uv", "args": ["run", "remote-ssh-mcp"], "enabled": true } 4. Report tmux attach -t remote-ssh-mcp/oracle-vps.

8.3 OpenPortal
Web UI for Sessions/Projects: `:3000`, `bun openportal --directory ~/Work --hostname 127.0.0.1`.

8.4 opencode serve
Port 4096, Session Move/Rename, Worktrees.

8.5 remote-ssh-mcp
tmux sessions per host.

8.6 No Syncthing Constraint
Session continuity via `opencode.db` + Session IDs, not via file sync.

8.7 Check Port/Process Conflicts
```bash
lsof -i :4096
lsof -i :3000
```

PHASE 9 – Agent: Local Voice Stack on Client
9.1 Productive Path (Table)

Component	Role	Where?
voxtype	Push-to-Talk STT, OSD, everyday	Laptop
Groq whisper-large-v3-turbo	Primary STT (fast)	Cloud
Whisper.cpp (VPS 8081)	Fallback STT	VPS
Piper	TTS	VPS/Laptop

9.2 voxtype (Productive Path)
- Whisper small
- Push-to-Talk: CapsLock (daemon hotkey, configured in daemon itself)
- State file
- OSD
- `pause_media`
- Max. 60 s
- `voxtype record toggle` (CLI command for external bindings)

Config: `~/.config/voxtype/config.toml`
Hyprland toggle binding: see 9.17 (`SUPER+R`)
Status: `voxtype status`

9.3 omarchy-wake-word
Wake-Word service.

9.4 STT Priority
1. Groq (fast/accurate)
2. Local Whisper.cpp (fallback, 16 kHz mono, Highpass 80 Hz, prompt field)

9.5 TTS Text Cleaning
- Code blocks → "Code is in the text"
- Markdown/URLs/Emojis out
- `%` → "percent"
- Capping 600 characters

9.6 Pronunciation Map
`Bot` → `Bott`, `systemctl` → `System-Control`, `JSON` → `Jot Ess Oh Enn`.

9.7 Hardware Reality
This laptop has no NVIDIA GPU (Intel UHD 630 + AMD RX 460) → Vulkan/ROCm path, no CUDA.

9.8 ydotool
System service + input group + `YDOTOOL_SOCKET`. Test Wayland injection.

9.9 Alternatives
Vocalinux/waystt/vox/kloyce/speak-to-ai remain alternatives but are not the productive path.

9.10 Prompt: Whisper.cpp + Vulkan
> On Laptop: 1. sudo pacman -S whisper-cpp ggml-vulkan 2. Check Vulkan drivers (Intel: vulkan-intel, AMD: vulkan-radeon) 3. If not: install appropriate driver 4. whisper-cpp

--download-model base 5. Test with sample file. Important: ggml-vulkan is required on Arch. The official whisper-cpp package is NOT compiled with Vulkan.

9.11 Prompt: Piper TTS
> On Laptop: 1. yay -S piper-tts-bin piper-voices-en-us 2. echo "Hello, I am your local assistant." | piper-tts --model /usr/share/piper-voices/en/en_US/amy/medium/en_US-amy-medium.onnx --output_file /tmp/test.wav 3. aplay /tmp/test.wav. Report TTS status.

9.12 Prompt: Vocalinux
> 1. curl -fsSL https://raw.githubusercontent.com/jatinkrmalik/vocalinux/main/install.sh -o /tmp/vl.sh && bash /tmp/vl.sh --interactive 2. Installer detects hardware, asks for engine (Whisper.cpp, Whisper, VOSK), loads model, sets up desktop integration. 3. Start: vocalinux 4. Operation: Hold right Alt key, speak, release → Text in focused window. 5. Voice commands: "new line", "delete", "select all". Important: 100% offline, Wayland compatible, GPU via Vulkan.

9.13 Prompt: waystt + ydotool
> 1. sudo pacman -S ydotool waystt-bin 2. sudo usermod -aG input $USER 3. ydotool service: sudo tee /etc/systemd/system/ydotool.service << 'EOF' [Unit] Description=ydotoold - ydotool daemon [Service] ExecStart=/usr/bin/ydotoold --socket-path /tmp/.ydotool_socket --socket-perm 0666 Restart=on-failure [Install] WantedBy=multi-user.target EOF sudo systemctl enable --now ydotool.service 4. export YDOTOOL_SOCKET=/tmp/.ydotool_socket in ~/.bashrc 5. Configure waystt: mkdir -p ~/.config/waystt cat > ~/.config/waystt/.env << 'EOF' TRANSCRIPTION_PROVIDER=local WHISPER_MODEL=ggml-medium.bin ENABLE_AUDIO_FEEDBACK=true BEEP_VOLUME=0.1 EOF waystt --download-model 6. Hyprland binding: bind = SUPER, R, exec, pgrep -x waystt >/dev/null && pkill --signal SIGUSR1 waystt || (waystt --pipe-to /usr/bin/ydotool type --file - &). Important: User in input group, log in again.

9.14 Prompt: vox + Vosk
> 1. sudo pacman -S vosk-api python-vosk 2. git clone https://github.com/SDG-Den/SDG-OS ~/projects/sdgos 3. cd ~/projects/sdgos/sdgos/vox 4. ./install.sh 5. vox config 6. vox daemon. Wake-Word: "system command". Examples: "system command, open firefox"; "system command, volume up"; "system command, search for cats online". Important: Vosk is faster but less accurate than Whisper.

9.15 Prompt: speak-to-ai
> 1. Download AppImage from https://github.com/AshBuk/speak-to-ai/releases 2. chmod +x speak-to-ai-*.AppImage 3. sudo usermod -aG input $USER 4. ./speak-to-ai-*.AppImage. Features: System Tray, CLI mode, WebSocket API, Wayland + X11, Voice Typing or Clipboard mode.

9.16 Prompt: kloyce
> 1. git clone https://github.com/andrew-pynch/kloyce 2. Follow installation instructions (systemd + Hyprland binding) 3. Start daemon 4. Web dashboard: http://localhost:9876. Features: SQLite history, dictionary corrections, GPU monitoring, Web dashboard, Transcription Queue.

9.17 Hyprland Binding for voxtype (Toggle)
CapsLock Push-to-Talk is configured in the daemon itself (see 9.2). The following binding is an additional toggle mode via `SUPER+R`:
```ini
# ~/.config/hypr/hyprland.conf
bind = SUPER, R, exec, voxtype record toggle
bind = SUPER SHIFT, R, exec, voxtype record toggle --mode clipboard
```

9.18 Summary: What belongs on the Client
Component	Package	Purpose	Recommendation
STT (Dictation)	`whisper-cpp` + `ggml-vulkan`	Speech → Text, local on GPU	★★★★★
TTS	`piper-tts-bin` + `piper-voices-en-us`	Text → Speech, local	★★★★★
System-wide dictation	`vocalinux` (Installer)	Whisper + Piper + Injection, ready	★★★★★
Alternative (minimal)	`ydotool` + `waystt-bin`	Push-to-Talk, manually configured	★★★★★
Voice Commands	`vosk-api` + `vox`	System commands by voice	★★★★★
AppImage (portable)	`speak-to-ai`	Minimal, AppImage	★★★★★
Web Dashboard	`kloyce`	Rust, SQLite, Dashboard	★★★★★
Productive Path	`voxtype`	Push-to-Talk, OSD, everyday	★★★★★

PHASE 10 – Agent: MCP Server
10.1 Real State: 26 configured, 21 active
Active: cloudflare-api, cloudflare-docs, cloudflare-bindings, cloudflare-builds, cloudflare-observability (optional), forgejo, gdrive (rclone), supabase, playwright, fetch, memory, filesystem, system-health, project-memory, resend, cloudflare-ai, pdfgen, ffmpeg, obsidian, voice-connect, ytdlp, remote-ssh.
Deactivated (deliberate): brightbean, pireel, searxng, github, vercel.

10.2 MCP Diet
Only load needed servers per project. Global 26 cost prompt tokens.

10.3 Project Scope
Work has its own enabled mask. Global config only definitions.

10.4 mcp-auth.json + OAuth
Document Vercel/GitHub OAuth flows.

10.5 Tool Reality
- Resend: ~106 tools
- Supabase: `listTables`
- PDFgen: `get_account`, `list_templates`

- GDrive: `config_listremotes`
- ytdlp: `download_audio`
- voice-connect: `tts`, `stt`, `job_status` (own MCP under `~/projects/voice-connect`, npm package `voice-connect-mcp`)

10.6 npx Pinning
`npx -y` pinning + mise paths (ffmpeg/remote-ssh) instead of "somewhere in PATH".

10.7 Prompt: rclone RC Daemon (VPS, System Service)
> 1. /etc/systemd/system/rclone-rcd.service: [Unit] Description=Rclone Remote Control Daemon After=network.target [Service] Type=simple User=ubuntu ExecStart=/usr/bin/rclone rcd --rc-no-auth --rc-addr=localhost:5572 Restart=always RestartSec=10 [Install] WantedBy=multi-user.target 2. sudo systemctl daemon-reload 3. sudo systemctl enable --now rclone-rcd 4. curl http://localhost:5572/core/version

10.8 Prompt: All MCP Servers
> Add the following mcp section to `~/.config/opencode/opencode.json` (on BOTH machines):
```json
{
  "mcp": {
    "cloudflare-api": {
      "type": "remote",
      "url": "https://mcp.cloudflare.com/mcp",
      "enabled": true,
      "headers": {"Authorization": "Bearer {env:CLOUDFLARE_API_TOKEN}"}
    },
    "cloudflare-docs": {
      "type": "remote",
      "url": "https://docs.mcp.cloudflare.com/mcp",
      "enabled": true
    },
    "cloudflare-bindings": {
      "type": "remote",
      "url": "https://bindings.mcp.cloudflare.com/mcp",
      "enabled": true
    },
    "cloudflare-builds": {
      "type": "remote",
      "url": "https://builds.mcp.cloudflare.com/mcp",
      "enabled": true,
      "headers": {"Authorization": "Bearer {env:CLOUDFLARE_API_TOKEN}"}
    },
    "cloudflare-observability": {
      "type": "remote",
      "url": "https://observability.mcp.cloudflare.com/mcp",
      "enabled": true,
      "headers": {"Authorization": "Bearer {env:CLOUDFLARE_API_TOKEN}"}
```

```
},
"forgejo": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@ric_/forgejo-mcp"],
  "env": {
    "FORGEJO_URL": "http://localhost:3000",
    "FORGEJO_TOKEN": "{env:FORGEJO_READ_TOKEN}"
  },
  "enabled": true
},
"gdrive": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "rclone-mcp-server"],
  "env": {"RCLONE_URL": "http://localhost:5572"},
  "enabled": true
},
"vercel": {
  "type": "remote",
  "url": "https://mcp.vercel.com",
  "enabled": false
},
"github": {
  "type": "remote",
  "url": "https://api.githubcopilot.com/mcp/",
  "enabled": false,
  "headers": {"Authorization": "Bearer {env:GITHUB_PERSONAL_ACCESS_TOKEN}"}
},
"supabase": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "supabase-mcp"],
  "env": {
    "SUPABASE_URL": "{env:SUPABASE_URL}",
    "SUPABASE_SERVICE_ROLE_KEY": "{env:SUPABASE_SERVICE_ROLE_KEY}"
  },
  "enabled": true
},
"playwright": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@playwright/mcp"],
  "enabled": true
},
"fetch": {
  "type": "local",
  "command": "npx",
```

```
  "args": ["-y", "@modelcontextprotocol/server-fetch"],
  "enabled": true
},
"memory": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@modelcontextprotocol/server-memory"],
  "enabled": true
},
"filesystem": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@modelcontextprotocol/server-filesystem", "/home/ubuntu/projects"],
  "enabled": true
},
"system-health": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "system-health-mcp"],
  "enabled": true
},
"project-memory": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@kaaustubh/project-memory-mcp"],
  "enabled": true
},
"resend": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "resend-mcp"],
  "env": {
    "RESEND_API_KEY": "{env:RESEND_API_KEY}",
    "SENDER_EMAIL_ADDRESS": "{env:SENDER_EMAIL_ADDRESS}"
  },
  "enabled": true
},
"cloudflare-ai": {
  "type": "local",
  "command": "npx",
  "args": ["-y", "@puspoaditya/cloudflare-workers-ai-mcp"],
  "env": {
    "CLOUDFLARE_ACCOUNT_ID": "{env:CLOUDFLARE_ACCOUNT_ID}",
    "CLOUDFLARE_API_TOKEN": "{env:CLOUDFLARE_API_TOKEN}"
  },
  "enabled": true
},
"pdfgen": {
```

```
      "type": "local",
      "command": "npx",
      "args": ["-y", "@pdfgen/mcp"],
      "env": {"PDFGEN_API_KEY": "{env:PDFGEN_API_KEY}"},
      "enabled": true
    },
    "ffmpeg": {
      "type": "local",
      "command": "npx",
      "args": ["ffmpeg-mcp-server"],
      "enabled": true
    },
    "obsidian": {
      "type": "local",
      "command": "npx",
      "args": ["-y", "obsidian-mcp@2", "serve", "--vault", "notes=/home/ubuntu/obsidian-vault"],
      "enabled": true
    },
    "voice-connect": {
      "type": "local",
      "command": "npx",
      "args": ["-y", "voice-connect-mcp"],
      "enabled": true
    },
    "ytdlp": {
      "type": "local",
      "command": "npx",
      "args": ["-y", "@kevinwatt/yt-dlp-mcp"],
      "enabled": true
    },
    "remote-ssh": {
      "type": "local",
      "command": "uv",
      "args": ["run", "remote-ssh-mcp"],
      "enabled": true
    }
  }
}
```

PHASE 11 – Agent: Keepalive & Monitoring
11.1 Oracle Threshold
<20% in the 95th percentile over 7 days (CPU + Net + RAM).

11.2 Keepalive
25% Duty (60 s on / 180 s off), real busy-loop load.

```bash
#!/bin/bash
# ~/keepalive.sh
while true; do
  timeout 60s sh -c 'while :; do :; done' &
  curl -s -o /dev/null https://www.google.com
  curl -s -o /dev/null https://api.github.com
  sleep 180
done
```
Service restart after change.

11.3 Monitoring Stack
Details in Section 4.1.

11.4 Beszel Alarms via API
- Status/CPU/RAM/Disk/ContainerHealth/SystemFailed
- Webhook as shoutrrr URL
- Send test message

11.5 monitor.sh
```bash
#!/bin/bash
source ~/.config/opencode/env.sh
STATUS="VPS Status $(date '+%Y-%m-%d %H:%M')\n"
STATUS+="\nDocker:\n"
for dir in ~/projects/*; do
  if [ -f "$dir/docker-compose.yml" ]; then
    cd "$dir"
    RUNNING=$(docker compose ps --status running 2>/dev/null | wc -l)
    STATUS+="$(basename $dir): $RUNNING Container\n"
  fi
done
CPU=$(top -bn2 | grep 'Cpu(s)' | tail -1 | awk '{print $2}')
STATUS+="\nCPU: ${CPU}%"
STATUS+="\nRAM: $(free -h | awk '/Mem:/ {print 3"/"2}')"
STATUS+="\nDisk: $(df -h / | awk 'NR==2 {print $5}')"
curl -s -X POST "https://api.telegram.org/bot
$TELEGRAM_BOT_TOKEN/sendMessage" -d"chat_id ={TELEGRAM_ADMIN_USER_ID}" -d
"text=${STATUS}"
```

11.6 update-check.timer
Weekly: only check/notify, not install.
- VPS: `apt list --upgradable`
- Client: `checkupdates`
Output via Telegram.

11.7 Log Rotation
journald cap + "0 failed units" as acceptance criteria.

PHASE 12 – Agent: Maintenance & Security
12.1 Prompt: Disk Cleanup
> 1. Create ~/projects/skills/disk-cleanup/SKILL.md: - First check largest directories with du -sh. - List candidates: docker system df, docker image prune -a --filter "until=168h", docker builder prune, apt clean, autoremove --purge, journalctl --vacuum-time=7d, /tmp older 7 days, npm cache clean --force, npm store prune, pip cache purge, uv cache clean. - ALWAYS show preview of files/sizes to be deleted. - Ask for confirmation. 2. npm install -g uncraft 3. Cron weekly: Preview via Telegram.

12.2 Prompt: System Health Monitoring
> 1. npm install -g system-health-mcp 2. In opencode.json under mcp: system-health with command npm, args ["-y","system-health-mcp"] 3. Test system-health-mcp --help 4. Ask for automatic monitoring via cron.

12.3 Prompt: Minimal Backup
> 1. ~/projects/backup/backup.sh: - ~/backups/daily/ - tar.zst for ~/projects, ~/.config/opencode, ~/.bashrc - delete local backups older than 24h 2. ~/projects/backup/sync-to-gdrive.sh: - rclone bisync ~/backups/daily/ gdrive-crypt:backups/ --resync (first time) - then --max-age 168h - delete files on Google Drive older than 7 days 3. systemd timer: backup.timer 03:00, gdrive-sync.timer 04:00 4. Activate both.

12.4 Prompt: Project Memory
> 1. npm install -g @kaaustubh/project-memory-mcp 2. In opencode.json under mcp: project-memory 3. In every project folder ~/projects/*/ an AGENTS.md with: - Project name, stack, run-commands - Important decisions (date + reason) - Learnings 4. Test: "What was the last decision for project X?" 5. Test: "Have we had this bug before?"

12.5 Three Backup Scripts
1. Projects/Configs (`tar.zst`, without `node_modules/.git/.venv`, without secrets)
2. Offsite-crypt (`rclone crypt` → Google Drive, Remote `gdrive-crypt:`)
3. Secrets-gpg (`gpg AES-256/SHA-512`)
+ Timer/Cron.

12.6 Secret Scan before Cloud Upload
`tar --exclude="env.sh"` doesn't work → `./env.sh`, `env.sh.*`, `env.sh.bak`.

12.7 gpg on Ubuntu
`--pinentry-mode loopback`.

12.8 find | while + pipefail
Loop kills script → use `PIPESTATUS`.

12.9 Retention
7/30/90 days. Restore test + `rclone check`.

12.10 Docker Prune without Volume Prune
```bash
docker image prune -a -f
docker builder prune -f
# NO docker volume prune
```

12.11 Disk Cleanup Skill
Preview + confirmation, uncraft.

12.12 Project Memory
Not just `AGENTS.md`: Initiatives, Check-in/Check-out, Decisions, Learnings, Preferences.

12.13 Btrfs
Scrub/Snapshots/fstrim as open operational point (currently 0 snapshots).

PHASE 13 – Agent: Analysis & Automation
13.1 Prompt: Playwright + yt-dlp
> 1. npm install -g @playwright/mcp 2. sudo apt install -y yt-dlp 3. npm install -g @kevinwatt/yt-dlp-mcp 4. In opencode.json under mcp: "playwright": {"type":"local","command":"npx","args":["-y","@playwright/mcp"]}, "ytdlp": {"type":"local","command":"npx","args":["-y","@kevinwatt/yt-dlp-mcp"]} 5. Test: "Navigate to example.com and take a screenshot" 6. Test: "Download the audio from https://youtube.com/watch?v=..."

13.2 Prompt: Research via E-Mail
> 1. npm install -g @chankamlam/resend-mcp 2. In opencode.json under mcp: resend with RESEND_API_KEY, SENDER_EMAIL_ADDRESS. 3. Skill ~/projects/skills/research-mail/SKILL.md: - Conduct research - Result as Markdown under ~/projects/research/YYYY-MM-DD-topic.md - Content as plain-text email to your address 4. Test with short research.

13.3 Prompt: Git Sync
> 1. ~/projects/git-sync/sync-all.sh: - Iterate over ~/projects/*/ with .git - git add -A; git commit -m "auto-sync $(date)"; git push - Log to ~/projects/git-sync/sync.log 2. Make executable. 3. systemd timer daily 23:00. 4. Test manually.

13.4 Scraping Limits
- No login
- CSRF/JS-Rendering
- Rate-Limits
- ToS
- Instagram Reel/Hashtag often not without login

13.5 yt-dlp
- Public videos: yes
- Instagram: no (CSRF)
- Facebook: only Open Graph with fallback

13.6 Git Sync
Only own repos (`easydoesit36/*`), skip foreign clones/empty repos, `gh auth git-credential` in timer.

13.7 Resend Domain Restriction
`curl` instead of `python-urllib` (Cloudflare 1010).

13.8 Research Mail Skill
Markdown local + E-Mail, PDF optional.

PHASE 14 – Agent: Creative Extensions
14.1 Prompt: Image Generation
> 1. npm install -g @puspoaditya/cloudflare-workers-ai-mcp 2. In opencode.json under mcp: cloudflare-ai with CLOUDFLARE_ACCOUNT_ID, CLOUDFLARE_API_TOKEN. 3. Test: "Generate an image of a fox in the snow"

14.2 Prompt: PDF Creation
> 1. npm install -g @pdfgen/mcp 2. In opencode.json under mcp: pdfgen with PDFGEN_API_KEY. 3. Test: "Create a PDF from this Markdown text"

14.3 Prompt: FFmpeg
> 1. npm install -g ffmpeg-mcp-server 2. In opencode.json under mcp: ffmpeg with command npx, args ["ffmpeg-mcp-server"]. 3. Test: "Cut the first 30 seconds from video.mp4" 4. Test: "Convert video.mov to mp4" 5. Test: "Remove silence from podcast.mp3"

14.4 Prompt: RTK Token Reduction
> 1. npm install -g rtk-mcp 2. In opencode.json under mcp: rtk with command npx, args ["-y","rtk-mcp"]. 3. Test: "Run 'ls -la' with RTK"

14.5 Prompt: Obsidian Vault
> 1. npm install -g obsidian-mcp@2 2. In opencode.json under mcp: obsidian with args ["-y","obsidian-mcp@2","serve","--vault","notes=/home/ubuntu/obsidian-vault"] 3. Test: "List all notes in my vault" 4. Test: "Search for notes on the topic 'Project'"

14.6 voice-connect
TTS/STT/Job-Status as MCP. Own MCP under `~/projects/voice-connect`, npm package `voice-connect-mcp`.

14.7 PDFgen
`get_account`, `list_templates`, Watermark in Free-Tier.

14.8 Cloudflare Workers AI
100k Requests/day, Flux 1 Schnell, Token with Workers-AI scope.

14.9 Optional Deactivated
Pireel Studio, BrightBean, SearXNG.

14.10 Obsidian Vault
`~/obsidian-vault` + `.obsidian`-Check.

14.11 FFmpeg MCP
`ffmpeg-static` bundled - check PATH.

14.12 RTK
Check `~/.config/rtk-mcp/config.toml`.

PART III – APPENDICES

Appendix A – Glossary
A.1 General
Term	Explanation
ACL	Access Control List - List of who can access what
Always Free	Permanently free cloud quota
AMD64	Architecture of Intel/AMD processors (x86_64)
API	Application Programming Interface - Interface for programs
API-Key	Secret key for an API
ARM64	Architecture of ARM processors (Ampere Altra)
Autarky	Independence from external systems
Availability Domain	Physically separated data center resources
Bootstrap	Basic installation of a system
CI/CD	Continuous Integration / Deployment - automated tests and release
Client	Machine that uses services
Cloud	Data center infrastructure over the internet
Cloudflare	CDN, DNS, Tunnel and Zero-Trust provider
Cloudflare Tunnel	Secure tunnel without open ports, for public services
Container	Isolated runtime environment
Cron-Job	Time-controlled task
Docker	Container platform
Egress	Outgoing data traffic
Firewall	Network filter
Forgejo	Self-hosted Git platform (Gitea Fork)
Funnel	Tailscale feature for public HTTPS URLs on `*.ts.net`
Grant	Modern Tailscale access rule
GPU	Graphics Processing Unit
Home-Region	Primary Oracle Cloud region

Idle-Reclamation	Oracle reclaims instances at <20% over 7 days
InfinityFree	Free web hosting service
IP Address	Unique network address
Kernel	Operating system core
Laptop	Mobile computer (Client)
MagicDNS	Tailscale device name resolution
Mesh-Network	Network with direct device connection
MCP	Model Context Protocol - Interface between AI and tools
Migration	Structured database change
Node.js	JavaScript runtime environment
npm	Node Package Manager
OCI	Oracle Cloud Infrastructure
OCPU	Oracle CPU unit
Omarchy	Arch-based Linux distribution
OpenCode	Terminal-based AI agent
OpenCode Zen	Free model gateway (no longer main path)
OpenRouter	Multi-provider API gateway
PAYG	Pay-As-You-Go
Phase	Section in the workflow
Plugin	Application extension
Port	Network service number
PostgreSQL	Open-source relational database
Prompt	Input to a model
Provider	AI model provider
Python	Programming language
Redis	In-memory database
Resend	E-Mail API with 3,000 emails/month free tier
Reverse-Proxy	Mediator between clients and backend
Runner	CI/CD execution process
Server	Service-providing machine
Serverprofis.de	German web host with free domain
Session	Running instance with state
SSH	Secure Shell - encrypted terminal connection
Stack	Combination of cooperating services
Swap	Memory area in case of RAM shortage
Tailscale	Mesh VPN based on WireGuard
Tailscale Funnel	Public access to Tailscale services via `*.ts.net`
Telegram	Messenger with Bot API
Timeout	Maximum wait time
tmux	Terminal multiplexer
TTFT	Time-to-First-Token
Ubuntu	Linux distribution based on Debian
uv	Python tool runner by Astral
VPS	Virtual Private Server
Wayland	Modern display protocol for Linux
WireGuard	Modern VPN protocol
Woodpecker CI	Self-hosted CI/CD engine
WordPress	Open-source CMS, installable on InfinityFree

x86_64	See AMD64
ydotool	Tool for text input under Wayland
Zero Trust	Security model without implicit trust

A.2 AI and Models
Term	Explanation
Agent	AI system for independent tasks
Agent-Loop	Cycle of thinking, tool call, observing
Chain	Ordered model list for fallback
Chat Completions API	Classic OpenAI API
Completion	Model-generated response
Fallback-Chain	See Chain
Gemini	Model family from Google
Gemini 2.5 Flash-Lite	Model with 30 RPM / 1,500 RPD
Context	Amount of text for model processing
Context Window	Maximum context size
LLM	Large Language Model
Muse Spark 1.2	OpenCode Zen model, 1M context
Nemotron 3 Ultra	Primary model, 1M context
Nemotron 3.5 Lightning	Secondary model, 30B MoE
NVIDIA NIM	NVIDIA Inference Microservices
NVIDIA API-Key	Format: `nvapi-...`
OpenRouter	Gateway to 500+ models
OpenRouter API-Key	Format: `sk-or-...`
OpenCode Zen API-Key	Sentinel `public` (anonymous)
Google AI Studio	Google's Gemini platform
Gemini API-Key	Format: `AIza...`
OpenAI Responses API	Newer OpenAI API (`v1/responses`)
opencode-auto-fallback	Fallback plugin with per-agent configuration
Cooldown	Skip time for faulty models
Exponential Backoff	Wait time doubling
Rate-Limit (RPM)	Requests per Minute
RPD	Requests per Day
TPM	Tokens per Minute
Reasoning	Multi-step thinking
Retryable Error	Repeatable error
Streaming	Continuous token output
Thinking	Model thinking mode
Token	Smallest text unit
Tool-Call	Call to external tools
User-Agent	Client identifier

A.3 Tools and Services
Term	Explanation
Cloudflared	Cloudflare Tunnel Daemon
Cloudflare MCP	MCP server for Cloudflare services

Conduit	Browser UI for OpenCode
disk-cleanup	Agent skill for safe storage cleanup
FFmpeg MCP	Video/Audio manipulation via FFmpeg
Forgejo Actions	GitHub Actions compatible CI/CD
Forgejo MCP	MCP server for Forgejo (`@ric_/forgejo-mcp`)
Groq	AI cloud with Whisper API
Home Assistant	Open-source smart home
InfinityFree	Free PHP/MySQL hosting with WordPress
kloyce	Rust-based STT tool with web dashboard
n8n	Open-source automation
Nextcloud	Self-hosted cloud storage
Obsidian MCP	Vault access for AI agents
OpenClaw	Persistent AI agent and gateway
OpenCode Live	Session synchronization
`@levra7/opencode-telegram-bot`	Telegram bot with multi-user, SSH, voice
PDFgen MCP	PDF generation from HTML/Markdown
Piper TTS	Local neural TTS
Pireel Studio	Browser-based video editor
Playwright MCP	Browser automation for E2E, Scrape, Screenshots
Postiz	Self-hosted social media scheduler
Project Memory MCP	Persistent project memory for AI agents
rclone	Cloud storage sync tool
rclone-mcp-server	MCP server for rclone RC API
remote-ssh-mcp	MCP for tmux+SSH sessions
Resend MCP	Email sending via Resend API
RTK-MCP	Token reduction for CLI outputs
speak-to-ai	AppImage for offline STT
Supabase	Open-source backend
System Health MCP	Real-time monitoring of CPU, RAM, Docker
Telegram Bot	Automated Telegram account
uncraft	Interactive Linux space recovery tool
Vercel MCP	Official MCP at `https://mcp.vercel.com`
Vocalinux	System-wide voice input
vox	Voice command system with wake word
waystt	Wayland-native speech-to-text tool
Whisper	Open-source speech recognition
Whisper.cpp	C++ port of Whisper
Woodpecker CI	Self-hosted CI/CD
yt-dlp MCP	YouTube download and transcription

A.4 Configuration
Term	Explanation
`.bashrc`	Bash configuration file
`.env`	Environment variable file
`tmux.conf`	tmux configuration file
`opencode.json`	OpenCode main configuration

`fallback.json`	Fallback plugin configuration (optional; replaced by opencode-auto-fallback)
`tui.json`	TUI configuration
`AGENTS.md`	Project memory file for AI agents
`issues.json`	Bug history for project memory
`~/.config/opencode/`	OpenCode configuration directory
`~/.local/share/opencode/`	OpenCode data directory
`opencode.db`	SQLite session database
`cooldownMs`	Cooldown in ms
`maxRetries`	Maximum retries
`timeoutMs`	Timeout in ms
`CLOUDFLARE_API_TOKEN`	Cloudflare API token
`CLOUDFLARE_ACCOUNT_ID`	Cloudflare Account ID
`CLOUDFLARE_TUNNEL_TOKEN`	Cloudflare Tunnel token
`CLOUDFLARE_DOMAIN`	Your domain
`OPENROUTER_API_KEY`	OpenRouter API key
`NVIDIA_API_KEY`	NVIDIA NIM API key
`GEMINI_API_KEY`	Google AI Studio API key
`GROQ_API_KEY`	Groq API key
`ZAI_API_KEY`	z.ai API key
`SUPABASE_URL`	Supabase URL
`SUPABASE_SERVICE_ROLE_KEY`	Supabase Service Role Key
`VERCEL_TOKEN`	Vercel API token
`GITHUB_PERSONAL_ACCESS_TOKEN`	GitHub PAT
`TELEGRAM_BOT_TOKEN`	Telegram Bot token
`TELEGRAM_ADMIN_USER_ID`	Admin Telegram ID
`RESEND_API_KEY`	Resend API key
`SENDER_EMAIL_ADDRESS`	Sender email address
`FORGEJO_READ_TOKEN`	Forgejo Read token
`FORGEJO_WRITE_TOKEN`	Forgejo Write token
`PDFGEN_API_KEY`	PDFgen API key
`YDOTOOL_SOCKET`	ydotool socket path

A.5 Platforms and Network
Term	Explanation
Ampere A1	ARM platform from Oracle
`VM.Standard.A1.Flex`	Oracle ARM instance shape
`oci-arm-catcher`	Poller for Oracle ARM capacity
Secure Boot	UEFI feature for signed bootloaders
Systemd-Boot	Bootloader from Omarchy
UEFI	Modern BIOS
Vulkan	Graphics API for GPU acceleration
Egress	Outgoing data traffic
Ingress	Incoming data traffic
Port 4096	Standard port for `opencode serve`
Port 4007	Standard port for Postiz
Port 3000	Standard port for Forgejo

Port 5678	Standard port for n8n
Port 8000	Supabase Envoy/Kong (Woodpecker optional on 8002)
Port 8002	Woodpecker CI (optional, to avoid collision with Supabase)
Port 8080	Temporal UI
Port 8081	Whisper Server
Port 5572	Standard port for rclone RC
Port 41641	Standard port for Tailscale
Subnet	Network sub-area
UDP	Connectionless protocol
VPN	Virtual Private Network
WireGuard	Modern VPN protocol
DNS	Domain Name System
Nameserver	DNS server for domain delegation
PipeWire	Audio server for Linux

A.6 Errors and Diagnosis
Term	Explanation
401	Unauthorized
403	Forbidden
404	Not Found
408	Request Timeout
429	Rate Limit
5xx	Server Error (500, 502, 503, 504)
FreeUsageLimitError	OpenCode Zen rate limit error
MissingSessionID	OpenCode Zen UA gate error
Out of host capacity	Oracle ARM capacity error
Auth-Error	Authentication error (401, 403)
Message-Length-Error	Context window exceeded
No-Response-Timeout	No response from model
TTFT-Timeout	No token within timeout
User-Abort	Cancelled by user

A.7 Social Media
Term	Explanation
Human-in-the-Loop	Human approval before publication
Draft	Draft, not yet published
Queue	Queue for publication
Hashtag	Keyword marked with #
Caption	Image caption or post text
Reel	Instagram short video
Story	Instagram Story (visible for 24h)
Carousel	Instagram post with multiple images
Shorts	YouTube short video
Hook	First seconds of a video
Trend-Sound	Currently popular audio clip
Alt-Text	Description for accessibility

Tagging	Linking other users
Engagement Rate	Interaction rate
Reach	Reach
Impressions	Impressions
Evergreen	Timeless reusable content

A.8 Voice and Audio
Term	Explanation
STT	Speech-to-Text
TTS	Text-to-Speech
Vosk	Lightweight speech recognition
Whisper	Open-source speech recognition by OpenAI
Whisper.cpp	C++ port of Whisper
Piper	Local neural TTS
Vocalinux	System-wide dictation tool
waystt	Wayland-native STT tool
vox	Voice command system
ydotool	Text input under Wayland
Wake-Word	Activation word
Push-to-Talk	Hold key to speak
Toggle-Mode	Press key to start/stop
Text-Injection	Inserting text into focused window
Vulkan	Graphics API for GPU acceleration
Voxtype	Push-to-Talk STT daemon
OSD	On-Screen Display
`tts_clean`	TTS text cleaning
Pronunciation-Map	Replacement rules for TTS

Appendix B – Troubleshooting
B.1 Phase 1 – Oracle VPS
Problem	Solution
Out of host capacity	Different Availability Domain, `oci-arm-catcher`
SSH fails	Check Security List (Port 22), public IP
Instance shut down	Check limits (max. 2 OCPUs/12 GB)
Docker Permission denied	`sudo usermod -aG docker $USER`, log in again
Node.js wrong version	Check `node -v` (22.x)
Idle-Reclamation	Set up Keepalive service (Phase 11)
Cloudflared not installed	`sudo apt install -y cloudflared`

B.2 Phase 2 – Laptop
Problem	Solution
Omarchy WLAN dead	`sudo rfkill unblock all`, `lspci -k`
Omarchy Touchpad dead	Kernel parameters in `/etc/default/grub`

Omarchy no audio	`sudo pacman -S pipewire pipewire-pulse wireplumber`
Docker Permission denied	`sudo usermod -aG docker $USER`, log in again
Tailscale doesn't connect	`sudo tailscale up --ssh` again

B.3 Phase 3 – OpenCode
Problem	Solution
`opencode models` shows no models	Check `opencode.json` JSON syntax
Fallback doesn't work	`fallback.json` to `enabled: true`
OpenRouter 401	`OPENROUTER_API_KEY` set? Check key on openrouter.ai/keys
OpenRouter 429	50 RPD reached – fallback takes over
OpenCode Zen 401	`apiKey: "public"` set? No account needed
OpenCode Zen Free-Promo ended	`curl -s https://opencode.ai/zen/v1/models -H 'authorization: Bearer public'`
NVIDIA NIM 401	`NVIDIA_API_KEY` set? Phone verification?
Gemini 401	`GEMINI_API_KEY` set?
Cooldown too long	Reduce `cooldownMs`
Timeout too short	Increase `timeoutMs`
Per-Agent-Fallback doesn't work	Check agent names in `fallback.json`

B.4 Phase 4 – Cloudflare + Tailscale
Problem	Solution
Cloudflared login fails	Open browser, call URL manually
Tunnel doesn't start	`sudo systemctl status cloudflared`
DNS route missing	`cloudflared tunnel route dns oracle-vps <hostname>`
Domain not reachable	Check NS delegation: `dig NS your-domain.de +short`
Quick Tunnel URL changes	Use Named Tunnel with own domain
Cloudflare MCP 401	Check `CLOUDFLARE_API_TOKEN`, check scopes
Tailscale Grants wrong	Check policy syntax (JSON)
Grants block SSH	Check `src`, `dst`, `ip` fields
Funnel not available	Set Funnel attribute in ACL (`nodeAttrs` → `attr: ["funnel"]`)
Funnel URL 404	Check if service runs locally: `curl http://localhost:4007`
Funnel limit reached	Max. 3 Funnel ports per node in Free Tier

B.5 Phase 5 – VPS Services
Problem	Solution
Docker Image not ARM64	`docker manifest inspect <image>`
Postiz doesn't start	`JWT_SECRET` min. 48 chars
Postiz TikTok OAuth bug	Use Production key
Postiz OAuth "Invalid state"	Disable third-party cookie blockers, check domain consistency
Postiz Temporal Silent Failure	Monitor Temporal status, health checks
Supabase doesn't start	`env` complete? Check logs
n8n container stops immediately	`stdin_open: true`, `tty: true`
n8n Merge-Node loses records	Regular log checks, monitoring
OpenClaw Gateway doesn't start	`systemctl --user status openclaw-gateway.service`

| OpenClaw ARM performance | Activate Node compile cache |

B.6 Phase 6 – Git + CI/CD
Problem	Solution
Forgejo doesn't start	`docker compose logs`, check DB connection
Forgejo Runner doesn't connect	Token, `journalctl -u forgejo-runner -f`
Forgejo Actions failed	Check labels
Woodpecker Agent doesn't connect	Check `WOODPECKER_AGENT_SECRET`
Forgejo MCP 401	Check `FORGEJO_READ_TOKEN`
Forgejo MCP Write-Tools missing	`FORGEJO_WRITE_TOKEN` set?

B.7 Phase 7 – Telegram
Problem	Solution
Bot doesn't respond	Check `TELEGRAM_ADMIN_USER_ID`
OpenCode Server not available	`opencode serve` running? URL correct?
Voice message without transcription	STT variables, Whisper Server
Whisper.cpp slow	`base` model, ARM64 NEON
systemd service doesn't start	`journalctl --user -u opencode-telegram-bot -f`

B.8 Phase 8 – Full Session Control
Problem	Solution
opencode-live doesn't sync	Same `opencode.db`, check Syncthing
remote-ssh-mcp doesn't connect	tmux 3.0+, python3, uv
Conduit Port occupied	`lsof -i :4096`
tmux Session gone	`tmux ls`, `tmux attach -t opencode`

B.9 Phase 9 – Local Voice Stack
Problem	Solution
whisper-cpp no GPU	`ggml-vulkan` installed?
ydotool Permission denied	`sudo usermod -aG input $USER`, log in again
ydotool socket not found	Set `YDOTOOL_SOCKET=/tmp/.ydotool_socket`
Vocalinux doesn't start	Python 3.9+, GTK 3, `vocalinux` in terminal
Vocalinux Wayland Injection	IBus-based flows, check active method
waystt no sound	PipeWire running? `systemctl --user status pipewire`
vox Wake-Word not recognized	Vosk model loaded? Check `vox config`
ydotool service doesn't start	Use system service instead of user service
Text doesn't appear in window	Wayland: ydotool, X11: xdotool, check app
Kloyce dashboard not reachable	Open `http://localhost:9876` in browser
speak-to-ai AppImage doesn't start	`chmod +x`, check input group

B.10 Phase 10 – MCP Server
Problem	Solution
MCP Server doesn't appear	Check `opencode.json` JSON syntax

Supabase MCP 401	`SUPABASE_SERVICE_ROLE_KEY` set?
Vercel MCP 401	Run OAuth flow (`/mcp` in OpenCode)
GDrive MCP doesn't connect	Check `sudo systemctl status rclone-rcd`
rclone not authenticated	`rclone config` again, renew token
Filesystem MCP path wrong	Use absolute path
System Health MCP shows no data	Test `npx system-health-mcp` manually
Resend MCP 401	`RESEND_API_KEY` set?
Cloudflare AI MCP 401	`CLOUDFLARE_API_TOKEN` with Workers-AI scope
PDFgen MCP 401	`PDFGEN_API_KEY` set? Watermark in Free Tier
FFmpeg MCP doesn't find ffmpeg	`ffmpeg-static` is bundled – check PATH

B.11 Phase 11 – Keepalive & Monitoring
Problem	Solution
Keepalive too aggressive	Reduce `dd` count (1024 instead of 2048)
Telegram notification doesn't arrive	Check `TELEGRAM_BOT_TOKEN` and `TELEGRAM_ADMIN_USER_ID`
Cron job doesn't run	`crontab -l`, check path to `monitor.sh`

B.12 Phase 12 – Maintenance
Problem	Solution
uncraft doesn't start	Node.js 18+, `npm install -g uncraft`
Backup timer doesn't run	`systemctl list-timers`, `journalctl -u backup.service`
rclone bisync conflict	`rclone bisync --resync` on first run
Project Memory doesn't write	`AGENTS.md` present in project folder?

B.13 Phase 13 – Analysis
Problem	Solution
Playwright doesn't find browser	`npx playwright install chromium`
yt-dlp download fails	`yt-dlp -U` (Update), check format
Resend email doesn't arrive	Domain verified? `SENDER_EMAIL_ADDRESS` correct?
Git sync push fails	SSH key stored on Forgejo/GitHub?

B.14 Phase 14 – Creative
Problem	Solution
Cloudflare AI image generation failed	API token with Workers-AI permission
PDFgen Watermark	Free Tier adds watermark – upgrade needed
FFmpeg MCP doesn't find file	Use absolute path
RTK-MCP doesn't filter	Check `~/.config/rtk-mcp/config.toml`
Obsidian Vault not found	`.obsidian` directory present? Use absolute path

B.15 InfinityFree WordPress
Problem	Solution
WordPress installation fails	Use Softaculous, check PHP 8.3

No SSH access	InfinityFree offers no SSH – manage files via FTP
Slow performance	Free hosting has limited CPU time – use own VPS for production
Custom domain cannot be added	First create account with subdomain, then add domain as addon

B.16 New Cases Production State
Problem	Solution
Postiz 502 after boot	Start Temporal first, then Postiz
Postiz JWT Default	Set 48-character secret
Caddy 400 "HTTP request to an HTTPS server"	Explicit `http://host:port`
Caddy basic_auth "Unexpected next token"	Multiline
Beszel: first user only via `create-user`, then 403	Agent key = Hub public key
Beszel: `InitializeUserSettings` overwrites webhooks	Add via PATCH
Groq 400 `tool_use_failed`	Always send tools; `tool_choice:"none"` only NVIDIA
Groq 429	Escalate immediately, don't backoff
Resend 1010	`curl` with User-Agent
Resend without domain	Only `onboarding@resend.dev`, only owner
`tar --exclude="env.sh"`	Match `./env.sh`
gpg on Ubuntu	`--pinentry-mode loopback`
`find | while` + pipefail	Loop kills script
Ubuntu cron `CRON_TZ`	System-wide TZ
VPS GitHub token 401	No push; Laptop `gh auth git-credential`
Forgejo-MCP from Laptop dead	`tailscale serve --tcp 3000`
Disk 89%	Block-Volume plan, don't delete blindly
Docker ports open despite ufw	Check VCN + DOCKER-USER
`opencode --dir` no longer exists in v2	`cd` + `opencode run`
OpenCode Zen Free-Promo ended	Check models live

Appendix C – Cost Overview
Component	Cost	Free-Tier Limits
Domain (optional)	0 € to ~10.44 €/year	Cloudflare Registrar (at-cost) or included in hosting packages
Oracle Cloud VPS (ARM)	0 €	2 OCPU/12 GB, 200 GB Storage, 10 TB Egress
Cloudflare Tunnel (Quick)	0 €	`*.trycloudflare.com`, not persistent
Cloudflare Tunnel (Named)	0 €	Requires own domain
Cloudflare Zero Trust	0 €	50 users
Cloudflare MCP	0 €	100,000 Requests/day
Cloudflare Workers AI	0 €	100,000 Requests/day, Flux 1 Schnell
Tailscale	0 €	6 users, unlimited devices, Grants included
Tailscale Funnel	0 €	Stable `*.ts.net` URL, max. 3 ports per node
OpenRouter	0 €	20 RPM, 50 RPD (1,000 RPD with $10 top-up)
OpenCode Zen	0 €	Anonymous via `public`, no account, 100 Req/day
NVIDIA NIM	0 €	40 RPM per model
Gemini 2.5 Flash-Lite	0 €	30 RPM / 1,500 RPD / 1M TPM

InfinityFree	0 €	5 GB Storage, unlimited bandwidth, WordPress
Serverprofis.de (optional)	from ~1.89 €/month	Free domain, 100 email accounts, cPanel
Supabase Cloud	0 €	500 MB DB, 1 GB Storage
Vercel Hobby	0 €	100 GB Bandwidth, 1,000 Builds/month
GitHub Free	0 €	Unlimited Public Repos
Google Drive API	0 €	1 billion Requests/day
Resend	0 €	3,000 Emails/month, 100/day
Telegram Bot API	0 €	Unlimited
Whisper (local)	0 €	ARM64 NEON
Piper TTS	0 €	GPL-3.0
Vocalinux	0 €	GPL-3.0
waystt	0 €	MIT
vox	0 €	Open Source
ydotool	0 €	MIT
Postiz	0 €	AGPL-3.0
n8n	0 €	Self-hosted
OpenClaw	0 €	Open Source
Forgejo + Actions	0 €	Self-hosted
Forgejo MCP	0 €	MIT
Woodpecker CI	0 €	Apache-2.0
Pireel Studio	0 €	AGPL-3.0
rclone	0 €	MIT
uncraft	0 €	Open Source
system-health-mcp	0 €	MIT
project-memory-mcp	0 €	MIT
@playwright/mcp	0 €	Apache-2.0
@kevinwatt/yt-dlp-mcp	0 €	MIT
@puspoaditya/cloudflare-workers-ai-mcp	0 €	MIT
@pdfgen/mcp	0 €	MIT
ffmpeg-mcp-server	0 €	MIT
rtk-mcp	0 €	MIT
obsidian-mcp	0 €	MIT
Freebuff	0 €	Ad-financed, passive on Client
speak-to-ai	0 €	MIT
kloyce	0 €	MIT
All MCP Servers	0 €	Open Source
Groq Free Tier	0 €	8000 TPM (only Small Model)
z.ai Free	0 €	limited
LM Studio/Ollama	0 €	local
Uptime Kuma	0 €	self-hosted
Beszel	0 €	self-hosted
Diun	0 €	self-hosted
Dozzle	0 €	self-hosted
cAdvisor	0 €	self-hosted
Caddy	0 €	self-hosted
fail2ban	0 €	Open Source
7zip/gpg	0 €	Open Source
Total	**0 €/month**	**+ optional 5-10 €/year for domain**

C.1 Workflow Overview
Phase	Who	Duration	Result
0	Human	60 Min	Accounts + optional Domain + all Keys
1	Human	20 Min	VPS running, Tailscale (Cloudflared optional)
2	Human	30 Min	Laptop running, Tailscale connected, Freebuff passive
3	Human	15 Min	OpenCode + 5 AI Providers working
4	Agent	-	Tailscale Funnel (Cloudflare optional) + MCP + Grants
5	Agent	-	Postiz, Supabase, n8n, OpenClaw, Social Media Workflow, HITL
6	Agent	-	Forgejo + CI/CD + MCP
7	Agent	-	Telegram Bot + Whisper + Piper
8	Agent	-	tmux, opencode-live, Conduit, remote-ssh-mcp
9	Agent	-	Local Voice Stack (Whisper.cpp, Piper, Vocalinux, waystt, vox)
10	Agent	-	rclone RC + All MCP Servers
11	Agent	-	Keepalive + Monitoring + Self-Maintenance
12	Agent	-	Storage Cleanup + Health + Backup + Memory
13	Agent	-	Playwright + yt-dlp + E-Mail + Git-Sync
14	Agent	-	Imagegen + PDF + FFmpeg + RTK + Obsidian

C.2 Decision Matrix for Phase 4
Your Situation	Recommended Phase(s)
No domain, no Cloudflare usage	4.1 → 4.5 (Funnel) → 4.4 (Grants)
No domain, Cloudflare desired	4.1 → 4.2 (Quick Tunnel) → 4.3 → 4.4
Own domain, Cloudflare desired	4.1 → 4.2 (Named Tunnel) → 4.3 → 4.4
Own domain, private access only	4.1 → 4.3 (optional) → 4.4
Both (public + private)	4.1 → 4.2 or 4.5 → 4.3 → 4.4

Human effort total: ~125 minutes (basic setup) + Agent phases 4-14.

Appendix D – Platform-Specific Social Media Adaptation
Platform	Format	Special Features
Instagram	Feed, Reel, Story, Carousel	Hashtags (max. 30), tags, Alt-Text, best posting times: 11-13 h, 19-21 h
TikTok	Video (9:16)	Hook in first 3 seconds, Trend sounds, Caption with keywords, no external links in text
YouTube Shorts	Video (9:16), max. 60 seconds	Title with keywords, Description, Tags
LinkedIn	Text + Image/Video	Professional tone, fewer hashtags, Document Carousel
X/Twitter	Text + Media	Short, concise, Threads possible

D.1 Platform Reality
- Instagram/TikTok OAuth needs Production key, not a "one click".
- Postiz-Temporal dependency.
- Draft → Queue → Publish as status chain.

- Human-in-the-Loop: signed links one-time + 24h timeout, then Telegram reminder, then E-Mail, then Draft.

Appendix E – Freebuff as a Passive Client Tool
Freebuff is a free CLI coding agent financed through terminal advertising. The advertising appears between agent turns, not in the generated code.

Positioning in the system: Freebuff is installed passively on the Omarchy client. It is not triggered by n8n, OpenClaw, or an automation step. It is not part of the workflow. It is a manually callable tool, comparable to git or ripgrep.

Installation:
```bash
npm install -g freebuff
```

No integration in: n8n workflows, OpenClaw skills, CI/CD pipelines, automated processes.

Properties: DeepSeek v4, MiMo 2.5 Pro, GLM 5.2, Minimax M3, plus Gemini 3.1 Flash Lite for file-finding. Nine specialized subagents (code-reviewer, browser-use, file-picker, etc.). Review engine at senior level, code generation falls back to DeepSeek V4 Flash.

Security notes: Only on Client, not on VPS. Do not connect sensitive project GitHub repos. Use as a pure review tool. Do not integrate into CI/CD or n8n.

Appendix F – Explanations for Beginners
F.1 What is a Terminal?
A terminal is a window where you type commands. Instead of clicking buttons, you write text. Example: `ls` shows you all files in the current folder.

F.2 What is SSH?
SSH (Secure Shell) is an encrypted connection to another computer. You type `ssh oracle-vps` and you are on the server – as if you were sitting in front of it.

F.3 What is Docker?
Docker is like an apartment for programs. Each program gets its own apartment (container) with everything it needs. This way programs cannot interfere with each other.

F.4 What is an API?
An API is an interface. Imagine you order in a restaurant. The kitchen is the API – you say what you want and get it. You don't need to know how the kitchen works.

F.5 What is a Repository?

A repository (repo) is a folder for code managed by Git. Git remembers every change. You can go back to an old version at any time.

F.6 What is a Commit?
A commit is a save point. You make a commit when you have finished a change. Git remembers: "At this point in time, the code looked like this."

F.7 What is a Branch?
A branch is a fork. You can try out a new feature without changing the main code. If it works, you merge the branch back.

F.8 What is an MCP Server?
MCP stands for Model Context Protocol. An MCP server is a mediator between AI and a service. It tells the AI: "You can use GitHub, here are the commands."

F.9 What is a Webhook?
A webhook is a reverse API call. Instead of you asking "Is there something new?", the service calls you: "Hey, there is something new!"

F.10 What is a Container?
A container is an isolated environment. It contains everything a program needs: code, libraries, settings. It runs the same everywhere.

F.11 What is a Token?
A token is a key. It is like a password, but only for programs. You store it in a file so you don't have to type it in every time.

F.12 What is a Systemd Service?
A systemd service is a program that starts automatically and runs in the background. It monitors itself and restarts if it crashes.

F.13 What is a Timer?
A timer is a time switch. It executes a task at a specific time. For example: "Backup at 3 AM."

F.14 What is DNS?
DNS (Domain Name System) is the telephone book of the internet. It translates `google.com` into an IP address like `142.250.185.46`.

F.15 What is TLS?
TLS (Transport Layer Security) is encryption on the internet. The lock symbol in the browser shows: This connection is secure.

F.16 What is Wayland?
Wayland is a modern display protocol for Linux. It replaces the old X11. Omarchy (Hyprland) uses Wayland. Some tools work differently under Wayland than under X11.

F.17 What is Vulkan?

Vulkan is a graphics API. It allows programs to use the GPU (graphics card) for computations. Whisper.cpp can be accelerated with Vulkan – this works with AMD, Intel, and NVIDIA.

F.18 What is a Wake-Word?
A wake-word is an activation word. You say "system command" and the computer knows: Now comes a command. Like "Hey Siri" or "Ok Google".

F.19 What is Text-Injection?
Text-Injection means: A program types text into another program. waystt or Vocalinux take your voice, convert it to text, and "type" it into the currently active window.

F.20 What is PipeWire?
PipeWire is an audio server for Linux. It manages microphone, speakers, and audio streams. waystt and other voice tools use PipeWire.

F.21 Tailnet vs. Internet
Tailnet is your private network (only your devices). Internet is public. Tailnet is more secure because no one from outside can access it.

F.22 Why 401 is a good sign
401 = "not authenticated". This means: The service is reachable but protected. Exactly right.

F.23 What a Reverse Proxy does
It accepts requests from outside and forwards them to the right service. Caddy does this with automatic TLS.

F.24 What a Healthcheck is
A test to see if a service really works (not just "runs"). Docker can automatically restart containers if the healthcheck fails.

F.25 What "Restore" means
Restoring a backup. A backup is only as good as the restore test.

F.26 Why Secrets don't belong in backups
Backups go to the cloud. Secrets in the cloud = leak risk.

F.27 What a Worktree is
A Git working directory. You can check out multiple branches at the same time.

F.28 What an Agent/Skill/Command is
- Agent: AI system for independent tasks
- Skill: Reusable capability (e.g., "Research + Mail")
- Command: Shortcut (e.g., `/voice on`)

Appendix G – Frequently Asked Questions (FAQ)

Question 1: What does this really cost?
Answer: 0 € for software and server. Optional ~10 €/year for your own domain. Cent amounts can occur with heavy AI usage.

Question 2: Do I need programming knowledge?
Answer: No. You copy the scripts and prompts. The AI executes them.

Question 3: How long does setup take?
Answer: About 3-4 hours for Phases 0-3 (manual). The AI does Phases 4-14.

Question 4: What happens if Oracle shuts down the server?
Answer: The Keepalive service prevents this. If it does happen, you can restart the server.

Question 5: Can I use this without Cloudflare?
Answer: Yes. Tailscale Funnel completely replaces Cloudflare.

Question 6: What if an AI API fails?
Answer: The fallback system automatically switches to the next provider.

Question 7: Can I use my own domain?
Answer: Yes. You just need to enter the nameservers at Cloudflare.

Question 8: How secure is the system?
Answer: Very secure. Tailscale encrypts everything, Zero Trust checks every device, no open ports.

Question 9: What happens to my data?
Answer: It stays on your server. Only texts are sent to the provider during AI usage.

Question 10: Can I extend the system?
Answer: Yes. Every phase is modular. You can add more MCP servers.

Question 11: What if I don't have a credit card?
Answer: Oracle requires one for verification but does not charge it. Alternatively, there are other cloud providers.

Question 12: How long are backups kept?
Answer: Locally 24 hours, on Google Drive 7 days.

Question 13: Can I use the system commercially?
Answer: Yes. The licenses (MIT, AGPL) allow commercial use. Freebuff is ad-financed.

Question 14: What if I need help?
Answer: The AI agent helps with errors. The troubleshooting in Appendix B covers the most common cases.

Question 15: Can I run the system on another server?
Answer: Yes. It runs on any Linux server with Docker.

Question 16: What is the difference between Server Voice and Client Voice?
Answer: Server Voice (Whisper Server on VPS) is for Telegram voice messages. Client Voice (Vocalinux, waystt, voxtype) is for dictation in applications on your laptop.

Question 17: Why do I need `ggml-vulkan` on Arch?
Answer: The official whisper-cpp package is not compiled with Vulkan. `ggml-vulkan` activates GPU acceleration.

Question 18: What is the difference between Vocalinux and waystt?
Answer: Vocalinux is a complete suite with tray, hotkeys, and installer. waystt is a minimal CLI tool that you configure yourself.

Question 19: Can I use voice commands without cloud?
Answer: Yes. vox uses Vosk locally, no cloud needed.

Question 20: What if ydotool doesn't work?
Answer: Check if you are in the input group and if the service is running. On Arch, the system service is often more reliable than the user service.

Question 21: Why no domain/no Cloudflare?
Answer: Tailnet is sufficient, less attack surface. Cloudflare is optional.

Question 22: Why no auto-update?
Answer: Production stack. Diun only reports; decisions and testing are manual.

Question 23: Why voxtype instead of Vocalinux?
Answer: Everyday suitable, Push-to-Talk, OSD. Vocalinux remains an alternative.

Question 24: Why Groq STT?
Answer: 0.3-0.6 s instead of 6-9 s. Local remains fallback.

Question 25: What if the VPS dies?
Answer: Backup/Restore. Laptop remains operational.

Question 26: What if the Laptop dies?
Answer: VPS continues to run. Secrets restore needed.

Question 27: Can I have multiple users?
Answer: Currently single-admin. Bot/Postiz/Forgejo need their own permissions.

Question 28: Is this GDPR compliant?
Answer: Document data flow, encrypt backups, set deletion periods.

Question 29: Why is Disk at 89%?
Answer: Images/Volumes. Block-Volume instead of deleting.

Appendix H – Learning Path for Students

H.1 Week 1 – Basics

Goal: Understand what the system does.

- Day 1: Read concept, look up terms in Appendix A
- Day 2: Terminal basics (`ls`, `cd`, `mkdir`, `nano`)
- Day 3: Understand SSH, connect to VPS
- Day 4: Docker basics (`docker ps`, `docker compose`)
- Day 5: First steps with OpenCode

H.2 Week 2 – Setup

Goal: Set up the system.

- Day 1: Phase 0 - Create accounts
- Day 2: Phase 1 - Bootstrap VPS
- Day 3: Phase 2 - Bootstrap Laptop
- Day 4: Phase 3 - Configure OpenCode
- Day 5: Phase 4 - Set up network

H.3 Week 3 – Services

Goal: Understand the services.

- Day 1: Phase 5.5 - Postiz
- Day 2: Phase 5.7 - n8n
- Day 3: Phase 5.8 - OpenClaw
- Day 4: Phase 5.9 - Social Media Workflow
- Day 5: Phase 5.10 - Human-in-the-Loop

H.4 Week 4 – Voice and Extensions

Goal: Extend the system.

- Day 1: Phase 6 - Git and CI/CD
- Day 2: Phase 7 - Telegram Bot
- Day 3: Phase 9 - Local Voice Stack (Vocalinux)
- Day 4: Phase 10 - MCP Servers
- Day 5: Phase 12 - Maintenance

H.5 Week 5 – Project Work

Goal: Implement your own project.

- Day 1-5: Implement your own project with the system (e.g., build a social media account, deploy your own website, build your own tool, configure voice commands)

H.6 Week 6 – Operations

- Read monitoring
- Test alarm
- Backup restore
- Rollback
- Incident Runbook
- Secret Scan

H.7 Week 7 – Security

- ufw/fail2ban
- SSH key-only
- Caddy Auth
- 401/000 Test
- Leak Rotation

H.8 Learning Resources
Topic	Resource
Terminal	`man ls`, `tldr ls`
SSH	`man ssh`, DigitalOcean Tutorials
Docker	docs.docker.com, Play with Docker
Git	git-scm.com/book, GitHub Skills
Python	python.org, Automate the Boring Stuff
JavaScript	javascript.info, MDN Web Docs
Linux	linuxjourney.com, Arch Wiki
Wayland	wayland.freedesktop.org
Voice	Vocalinux Docs, waystt GitHub

Appendix I – Local Voice Control Reference
I.1 Comparison of Voice Tools
Tool	Type	STT Engine	Wayland	GPU	Installation	Recommendation
Voxtype	Push-to-Talk Daemon	Whisper small	Yes	Vulkan/ROCm	Package/Config	Productive Path
Vocalinux	Complete Suite	Whisper.cpp, Whisper, VOSK	Yes	Vulkan	One Installer	Alternative
waystt	CLI Tool	Whisper (local/API)	Yes (ydotool)	-	AUR + ydotool	Alternative
ydotool	Injection Tool	-	Yes	-	pacman	Mandatory for Wayland Injection
vox	Voice Commands	Vosk	Yes	-	Git + install.sh	Alternative
speak-to-ai	AppImage	Whisper	Yes	-	AppImage	Alternative
kloyce	Rust + Dashboard	Whisper	Yes	nvidia-smi	Git	Alternative

I.2 Installation Commands at a Glance
```bash
# Whisper.cpp + Vulkan
sudo pacman -S whisper-cpp ggml-vulkan
whisper-cpp --download-model base

# Piper TTS
yay -S piper-tts-bin piper-voices-en-us
echo "Hello" | piper-tts --model
/usr/share/piper-voices/en/en_US/amy/medium/en_US-amy-medium.onnx --output_file
/tmp/test.wav
aplay /tmp/test.wav
```

```
# ydotool
sudo pacman -S ydotool
sudo usermod -aG input $USER
```

I.3 ydotool Service (Arch)
```bash
sudo tee /etc/systemd/system/ydotool.service << 'EOF'
[Unit]
Description=ydotoold - ydotool daemon
[Service]
ExecStart=/usr/bin/ydotoold --socket-path /tmp/.ydotool_socket --socket-perm 0666
Restart=on-failure
[Install]
WantedBy=multi-user.target
EOF
sudo systemctl enable --now ydotool.service
export YDOTOOL_SOCKET=/tmp/.ydotool_socket
```

I.4 Hyprland Binding for waystt
```ini
# In ~/.config/hypr/hyprland.conf
bind = SUPER, R, exec, pgrep -x waystt >/dev/null && pkill --signal SIGUSR1 waystt ||
(waystt --pipe-to /usr/bin/ydotool type --file - &)
bind = SUPER SHIFT, R, exec, pgrep -x waystt >/dev/null && pkill --signal SIGUSR1 waystt
|| (waystt --pipe-to wl-copy &)
```

I.5 Hyprland Binding for voxtype (Toggle)
CapsLock Push-to-Talk is configured in the daemon itself (see Phase 9.2). The following binding is an additional toggle mode via `SUPER+R`:
```ini
bind = SUPER, R, exec, voxtype record toggle
bind = SUPER SHIFT, R, exec, voxtype record toggle --mode clipboard
```

I.6 Vosk Model for vox
```bash
mkdir -p ~/.local/share/vox/models
cd ~/.local/share/vox/models
wget https://alphacephei.com/vosk/models/vosk-model-small-en-us-0.15.zip
unzip vosk-model-small-en-us-0.15.zip
```

I.7 Groq STT
- Model: `whisper-large-v3-turbo`
- Latency: 0.3-0.6 s

- Primary, local fallback

I.8 Piper TTS
- Voice: `en_US-amy-medium`
- Opus 48 kHz
- `tts_clean`
- Pronunciation map
- `voice on|off`

I.9 Voxtype
- Config: `~/.config/voxtype/config.toml`
- Hotkey: CapsLock (Push-to-Talk, configured in daemon)
- State file
- OSD
- `pause_media`
- Max. 60 s
- `voxtype record toggle` (CLI)

PART IV – ACCEPTANCE & TARGET DECISIONS

Acceptance Criteria (Definition of Done)
VPS
```bash
docker ps --format '{{.Names}}' | wc -l # 30
docker ps --filter health=unhealthy --format '{{.Names}}' # empty
systemctl --failed --no-legend | wc -l # 0
ss -lnt | grep -E ':(4007|8000)\b' # internal only
curl -s -o /dev/null -w '%{http_code}' http://127.0.0.1:3000 # 200/307
```

Tailnet UI without Auth
```bash
curl -s -o /dev/null -w '%{http_code}' http://100.90.252.42:8443 # 401
```

Backup
```bash
rclone check ~/backups gdrive-crypt: --one-way # 0 differences
```

Secrets
```bash
tar -tzf ~/backups/opencode-config-*.tar.zst | grep -E 'env\.sh|\.env$' # empty
```

These checks are part of every phase, not just "at the end".

Target Decisions
Decision	Reason
Tailnet-only, cloudflared inactive, no domain	Less attack surface, no costs, Zero Trust default
Oracle threshold <20% (95th percentile, 7 days)	Oracle raised from 15% to 20% in 2025
`env.sh` 600 + `{env:}` instead of `~/.bashrc`	No inline keys, secure interpolation
voxtype + wake-word as default	More everyday suitable than Vocalinux/waystt
Groq whisper-large-v3-turbo primary, local fallback	0.3-0.6 s instead of 6-9 s
Groq/NVIDIA/OpenRouter/local; Zen not main path	Zen Free-Tier ended
Forgejo Runner sufficient; Woodpecker optional	Fewer containers, one tool
opencode-auto-fallback + disabled_providers/whitelists	No `fallback.json` plugin needed anymore
Session IDs + opencode.db + Worktrees, no sync constraint	No Syncthing for sessions
Only 22/tcp public; everything else Tailnet/Loopback	No open ports

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Sources
- Live checks (Client + VPS)
- `~/projects/SYSTEM.md`
- `PORTS.md`
- `BACKUP.md`
- `memory/projects/vps.md`
- `backup.md`
- `conventions.md`

Nothing changed: no file, no config, no service.

End of Concept Version 6.0